

- 3.** A charged particle at rest is subjected to an external magnetic field $\vec{B}$. If no other force acts on the particle, then it will :
- (a) accelerate along $\vec{B}$
 - (b) remain at rest
 - (c) move in a circular path
 - (d) move with a constant speed along $\vec{B}$

5. A current carrying loop is placed in a uniform magnetic field. The torque acting on it does ***not*** depend upon the :

- | | |
|----------------------|-------------------------|
| (a) magnetic field | (b) current in the loop |
| (c) area of the loop | (d) shape of the loop |

- 6.** Two identical coils carrying equal currents are held concentric with their planes perpendicular to each other. The ratio of the magnitude of magnetic field at the centre of one coil to that of the resultant magnetic field at the centre is :

(a) $\frac{1}{\sqrt{2}}$

(b) $\sqrt{2}$

(c) $\frac{1}{2}$

(d) 2

7. An electron is revolving around a nucleus in a circular orbit of radius r with a constant speed v . The magnetic moment associated with the circulating current is :

(a) $\frac{evr}{4}$

(b) $4\ evr$

(c) $2\ evr$

(d) $\frac{evr}{2}$

- 32.** (a) (i) Two long parallel straight conductors carrying current I_1 and I_2 are kept r distance apart in air. Obtain an expression for the force per unit length on one conductor due to the magnetic field produced by the other conductor. Hence, define one ampere. Under what condition will the force between the conductors be attractive in nature ?
- (ii) A closely wound circular coil of radius 6.28 cm, having 50 turns carries a steady current of 4 A. Find the magnetic field at the centre of the coil. How will the magnitude of the magnetic field be affected if the radius of the coil is halved keeping other factors same ?

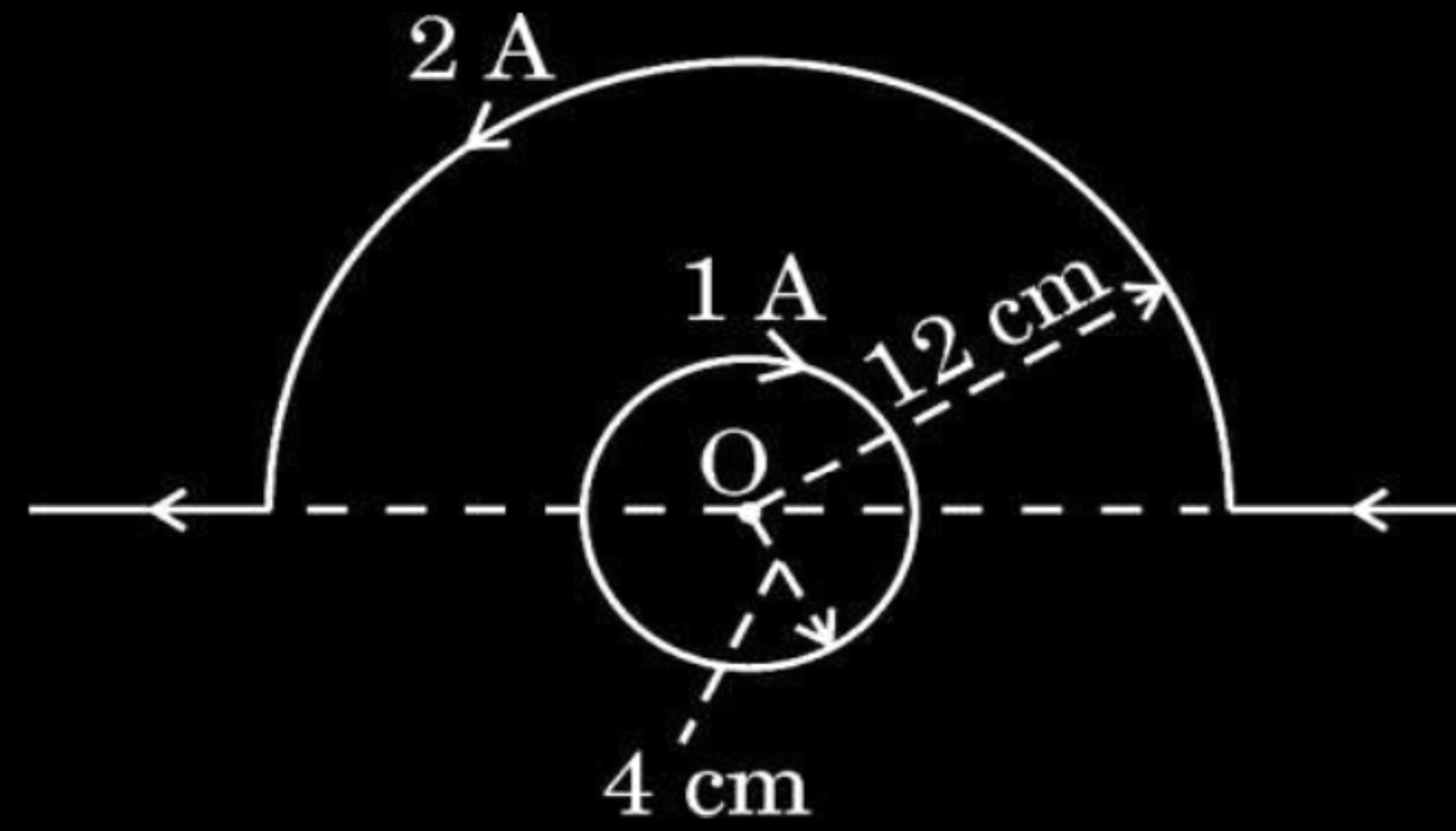
- (b) (i) A particle of mass m and charge q , at the origin moves with a velocity $\vec{v}$ in xy -plane making an angle θ with x -axis. It is subjected to a uniform magnetic field $\vec{B}$ along x -axis. Justify that the particle will move in a helical path. Hence, obtain expression for the radius of the helix.
- (ii) A long straight wire kept horizontally carries a current of 4 A from west to east direction. Find the direction and magnitude of magnetic field $\vec{B}$ at a point 20 cm below the wire.

18. *Assertion (A) :* A current carrying square loop made of a wire of length L is placed in a magnetic field. It experiences a torque which is greater than the torque on a circular loop made of the same wire carrying the same current in the same magnetic field.

Reason (R) : A square loop occupies more area than a circular loop, both made of wire of the same length.

- 23.** A particle of charge q and mass m starts moving with uniform velocity $v_0 \hat{i}$. Specify the direction of magnetic field which should be set up in the region so that the particle moves (a) straight undeviated, and (b) in a circle. Justify your answers.

27. A current carrying circular loop and a straight wire bent partly in the form of a semicircle are placed as shown in the figure. Find the magnitude and direction of net magnetic field at point O. 3



7. The ratio of the magnitude of magnetic moment to that of angular momentum, of an electron revolving around the nucleus in a hydrogen atom is :

(a) Zero

(b) $9.27 \times 10^{-24} \text{ C/kg}$

(c) $8.8 \times 10^{10} \text{ C/kg}$

(d) $6.6 \times 10^{-12} \text{ C/kg}$

29. Two infinitely long parallel conductors A and B, separated by 2 cm, carry 6 A and 2 A currents in opposite directions respectively. Find :

- (a) the net magnetic field at a point midway between A and B.
- (b) the force acting per unit length on B.

2. An electron enters a uniform magnetic field with speed v . It describes a semicircular path and comes out of the field. The final speed of the electron is :

(a) Zero

(b) v

(c) $\frac{v}{2}$

(d) $2v$

- 26.** Two circular loops A and B, each of radius 3 m, are placed coaxially at a distance of 4 m. They carry currents of 3 A and 2 A in opposite directions respectively. Find the net magnetic field at the centre of loop A.

18. *Assertion (A) :* When radius of a circular loop carrying a steady current is doubled, its magnetic moment becomes four times.

Reason (R): The magnetic moment of a circular loop carrying a steady current is proportional to the area of the loop.

- 22.** A wire of length L is bent round in the form of (i) a square, and then (ii) an equilateral triangle. If current I is passed through each of them, find the ratio of magnetic moment of the square loop to that of the triangle.

- 33.** (a) (i) Write the principle and explain the working of a moving coil galvanometer. A galvanometer as such cannot be used to measure the current in a circuit. Why ?
- (ii) Why is the magnetic field made radial in a moving coil galvanometer ? How is it achieved ?

- (b) (i) Derive an expression for magnetic field on the axis of a current carrying circular loop.

10. In a moving coil galvanometer, the deflecting torque τ acting on the coil is related to the current I flowing through it as :

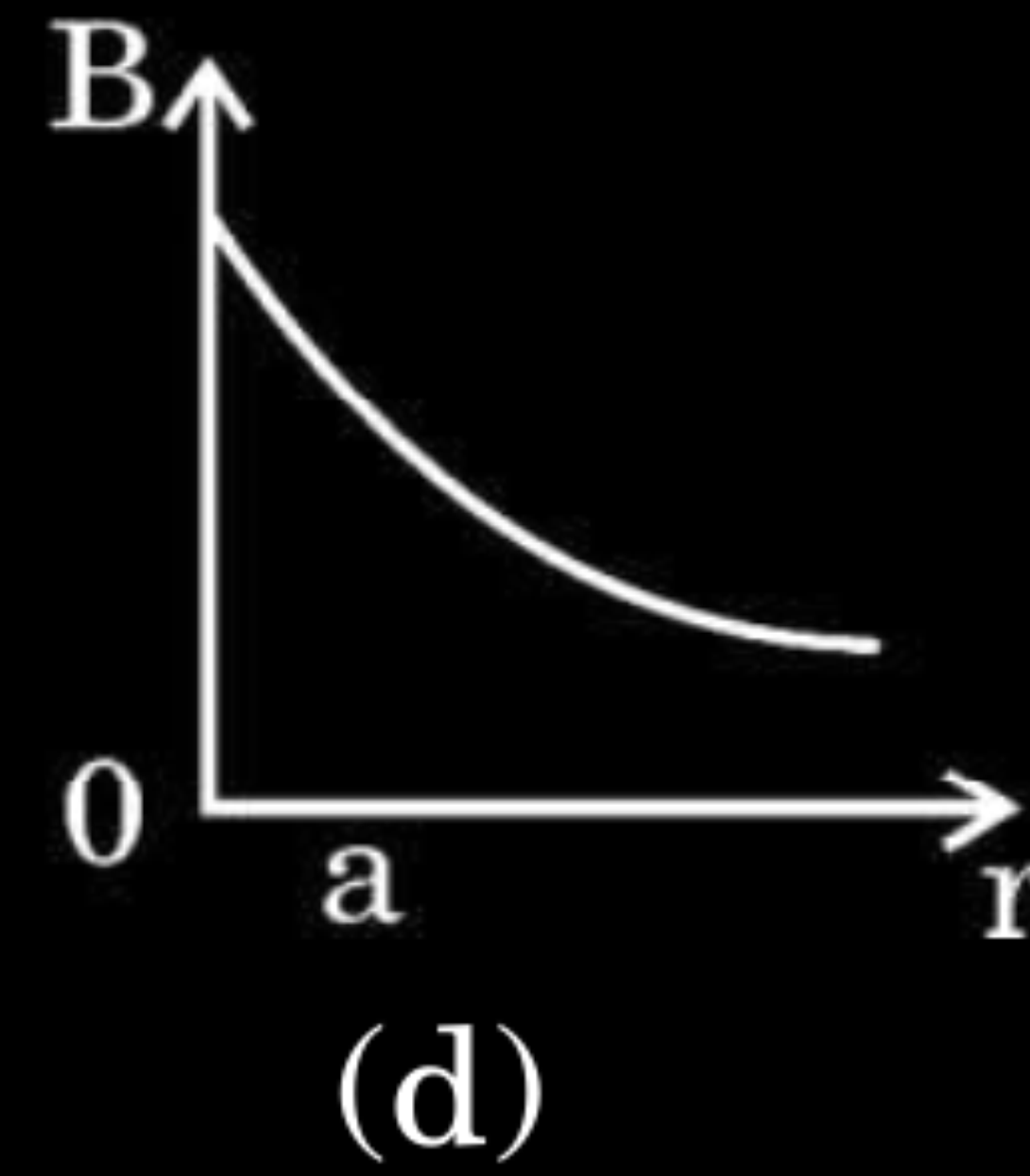
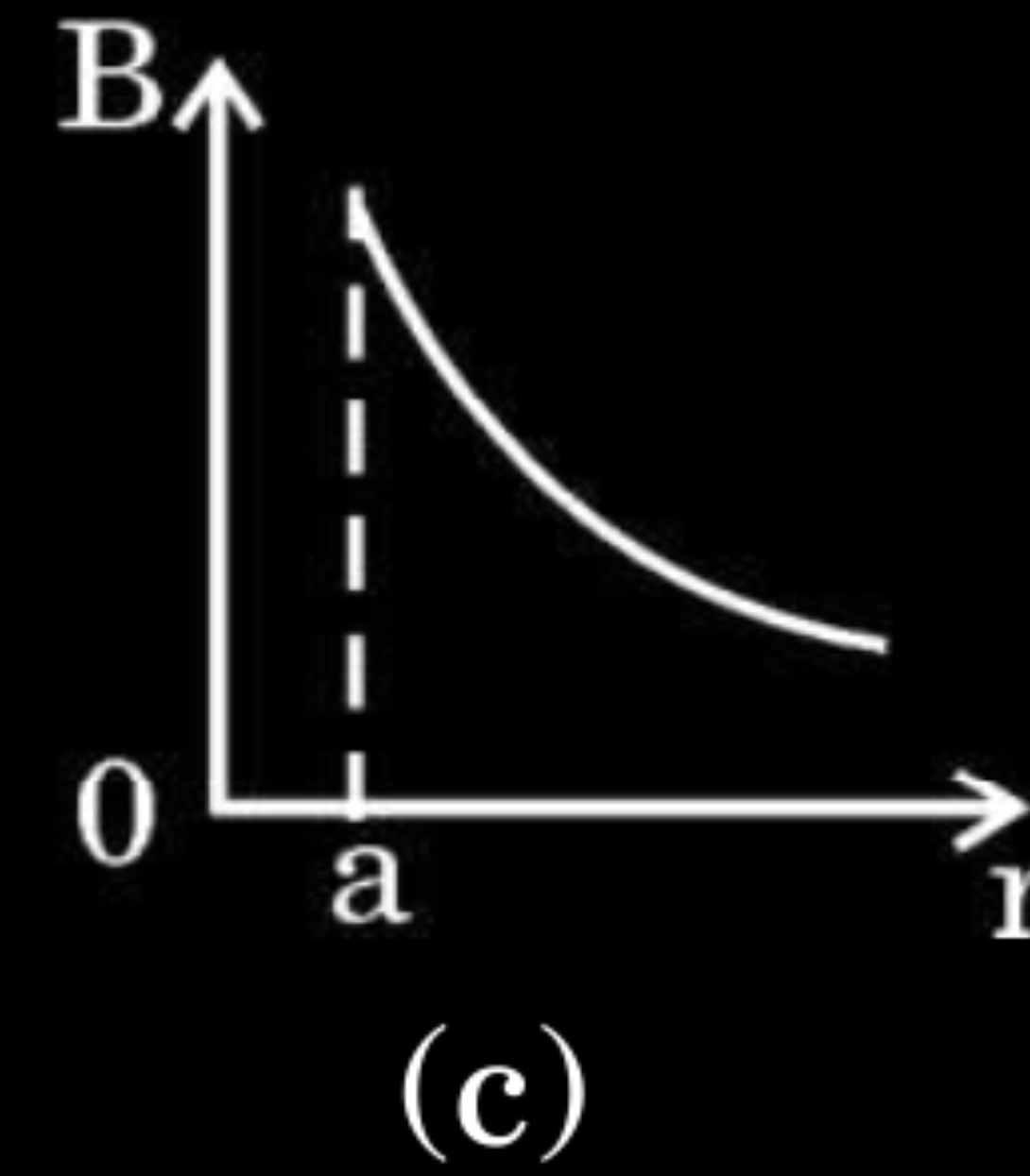
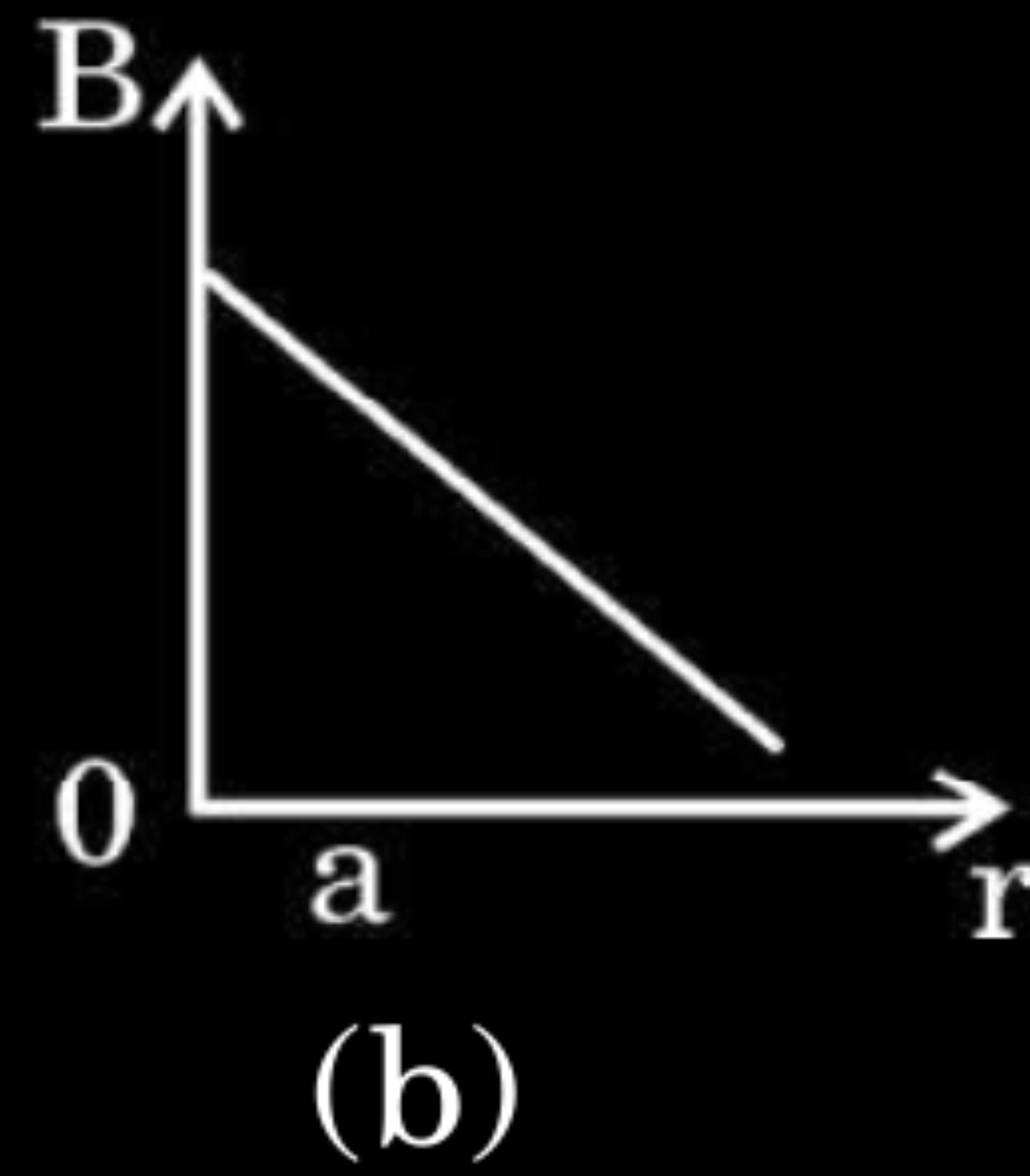
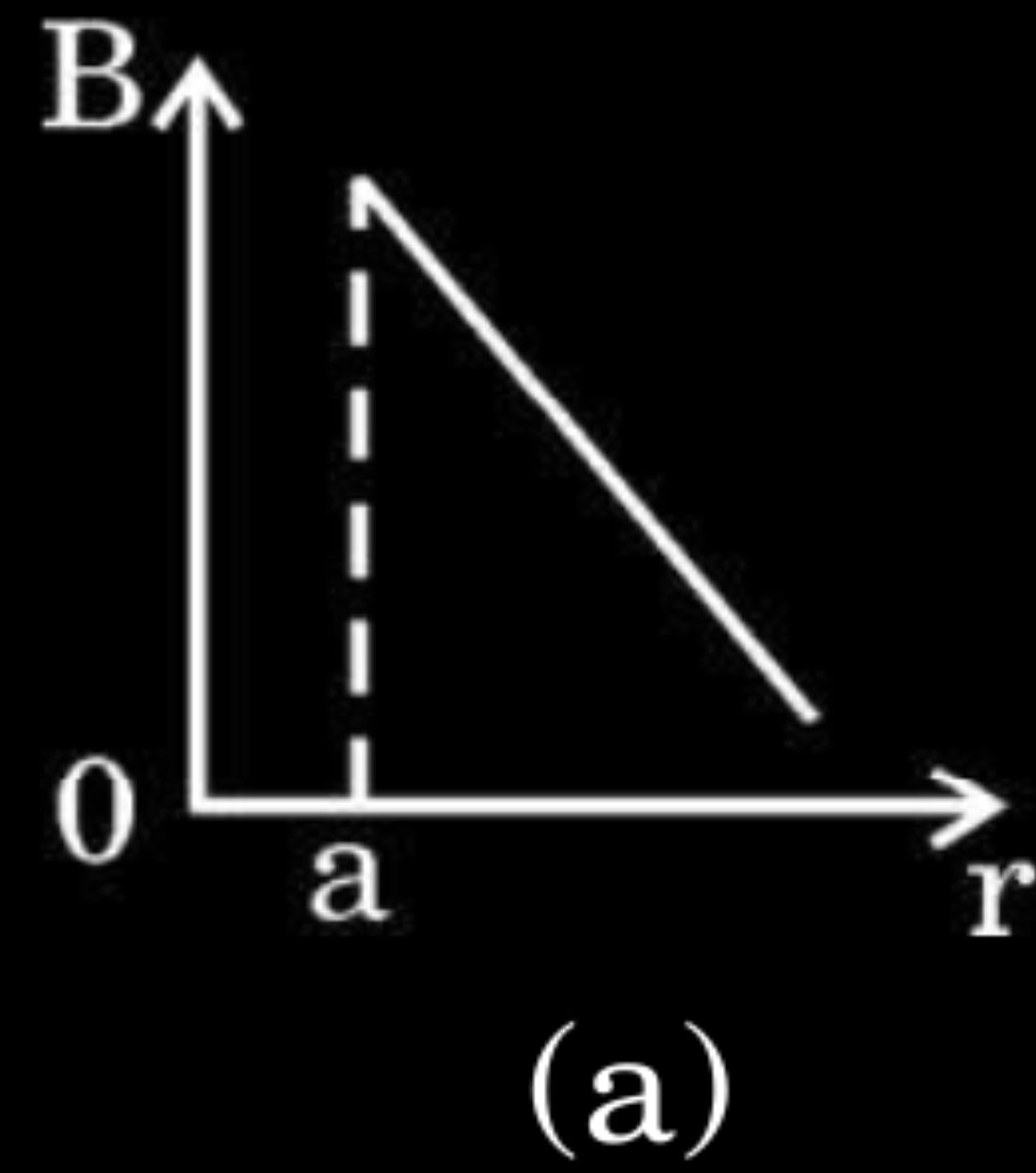
(a) $\tau \propto I^3$

(b) $\tau \propto I^2$

(c) $\tau \propto I$

(d) $\tau \propto \sqrt{I}$

15. Which of the following graphs correctly represents the variation of the magnitude of the magnetic field outside a straight infinite current carrying wire of radius 'a', as a function of distance 'r' from the centre of the wire ?



- 23.** Two wires of equal lengths are shaped in the form of a square loop and a circular loop. Both loops are suspended in a uniform magnetic field. Prove that for the same current, the circular loop will experience larger torque. 2

5. A particle of mass m and charge q moving with a uniform velocity $\vec{v} = v_{0x} \hat{i} + v_{0y} \hat{j}$ enters a region with a magnetic field $\vec{B} = B_0 \hat{j}$. After some time, an electric field $\vec{E} = E_0 \hat{j}$ is also switched on in the region. The resulting path described by the particle will be :
- (a) a circle in x-z plane
 - (b) a parabola in x-y plane
 - (c) a helix with constant pitch
 - (d) a helix with increasing pitch

- 19.** A wire of length l is in the form of a circular loop A of one turn. This loop is reshaped into loop B of three turns. Find the ratio of the magnetic fields at the centres of loop A and loop B for the same current through them.

- 21.** An alpha particle is projected with velocity $\vec{v} = (3.0 \times 10^5 \text{ m/s}) \hat{i}$ into a region in which magnetic field $\vec{B} = [(0.4 \text{ T}) \hat{i} + (0.3 \text{ T}) \hat{j}]$ exists. Calculate the acceleration of the particle in the region. $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x, y and z axis respectively and charge to mass ratio for alpha particle is $4.8 \times 10^7 \text{ C/kg}$.

- 33.** (a) (i) State Biot-Savart's law for the magnetic field due to a current carrying element. Use this law to obtain an expression for the magnetic field at the centre of a circular loop of radius 'a' and carrying a current 'I'. Draw the magnetic field lines for a current loop indicating the direction of magnetic field.
- (ii) An electron is revolving around the nucleus in a circular orbit with a speed of 10^7 m s^{-1} . If the radius of the orbit is 10^{-10} m , find the current constituted by the revolving electron in the orbit.

- (b) (i) Derive an expression for the force acting on a current carrying straight conductor kept in a magnetic field. State the rule which is used to find the direction of this force. Give the condition under which this force is (1) maximum, and (2) minimum.
- (ii) Two long parallel straight wires A and B are 2.5 cm apart in air. They carry 5.0 A and 2.5 A currents respectively in opposite directions. Calculate the magnitude of the force exerted by wire A on a 10 cm length of wire B.

1. A long straight wire of radius 'a' carries a steady current 'I'. The current is uniformly distributed across its area of cross-section. The ratio of magnitude of magnetic field $\vec{B}_1$ at $\frac{a}{2}$ and $\vec{B}_2$ at distance 2a is

1

(a) $\frac{1}{2}$

(b) 1

(c) 2

(d) 4

24. Briefly explain why and how a galvanometer is converted into an ammeter.

13. Beams of electrons and protons move parallel to each other in the same direction. They

1

- (a) attract each other.
- (b) repel each other.
- (c) neither attract nor repel.
- (d) force of attraction or repulsion depends upon speed of beams.

2. Two horizontal thin long parallel wires, separated by a distance r carry current I each in the opposite directions. The net magnetic field at a point midway between them, will be

1

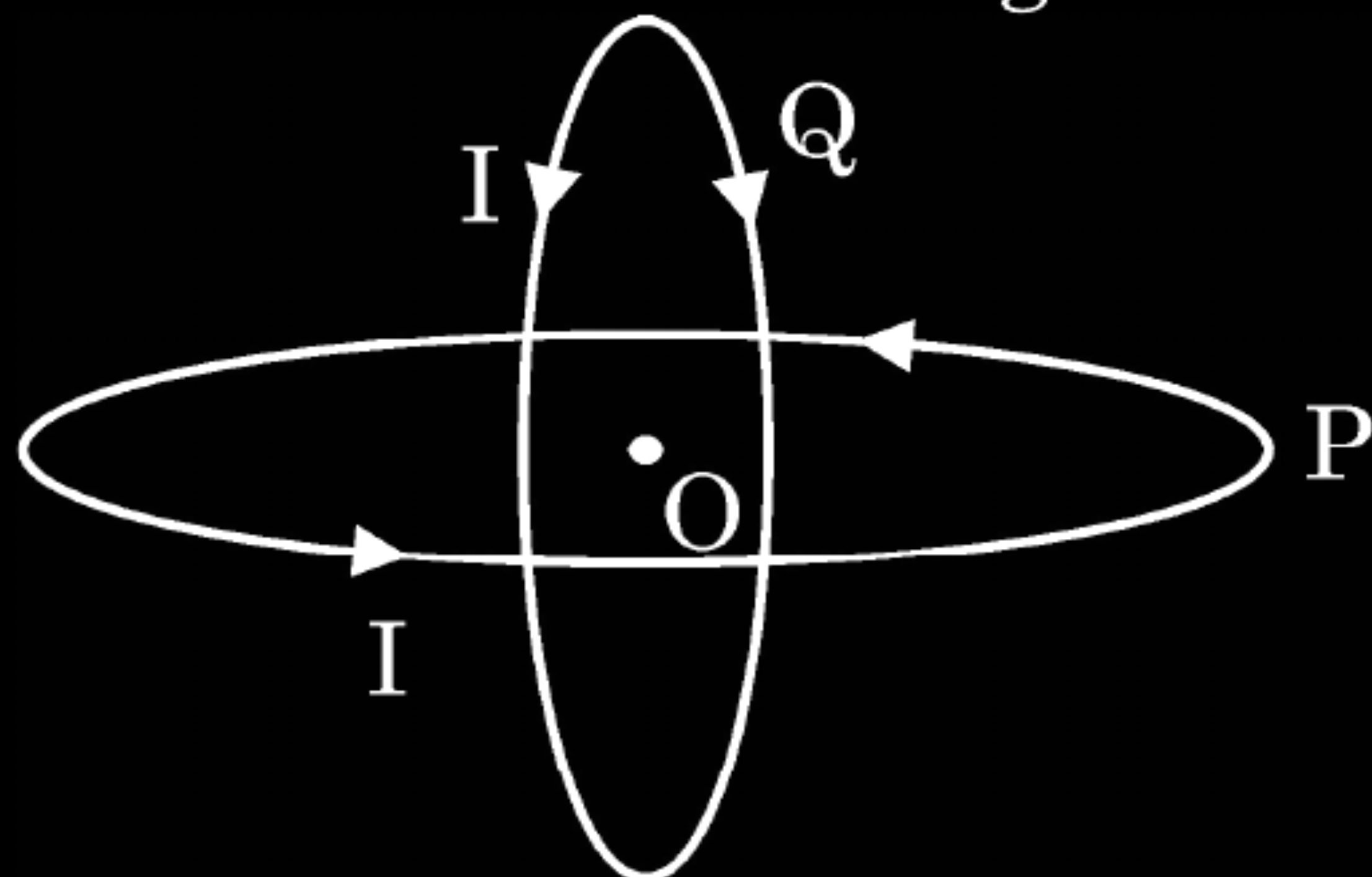
- (A) zero
- (B) $\left(\frac{\mu_0 I}{2\pi r}\right)$ vertically downward
- (C) $\left(\frac{2\mu_0 I}{r}\right)$ vertically upward
- (D) $\left(\frac{\mu_0 I}{\pi r}\right)$ vertically downward

18. **Assertion (A)** : The deflecting torque acting on a current carrying loop is zero when its plane is perpendicular to the direction of magnetic field.

1

Reason (R) : The deflecting torque acting on a loop of magnetic moment $\vec{m}$ in a magnetic field $\vec{B}$ is given by the dot product of $\vec{m}$ and $\vec{B}$.

23. (a) Two identical circular loops P and Q, each of radius R carrying current I are kept in perpendicular planes such that they have a common centre O as shown in the figure.



Find the magnitude and direction of the net magnetic field at point O. **2**

- (b) A long straight conductor kept along $X'X$ axis, carries a steady current I along $+x$ direction. At an instant t , a particle of mass m and charge q at point (x, y) moves with a velocity $\vec{v}$ along $+y$ direction. Find the magnitude and direction of the force on the particle due to the conductor.

29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.
- (b) A galvanometer shows full scale deflection for current I_g . A resistance R_1 is required to convert it into a voltmeter of range $(0 - V)$ and a resistance R_2 to convert it into a voltmeter of range $(0 - 2V)$. Find the resistance of the galvanometer.

2. Two long parallel wires kept 2 m apart carry 3A current each, in the same direction. The force per unit length on one wire due to the other is

1

(A) $4.5 \times 10^{-5} \text{ Nm}^{-1}$, attractive

(B) $4.5 \times 10^{-7} \text{ N/m}$, repulsive

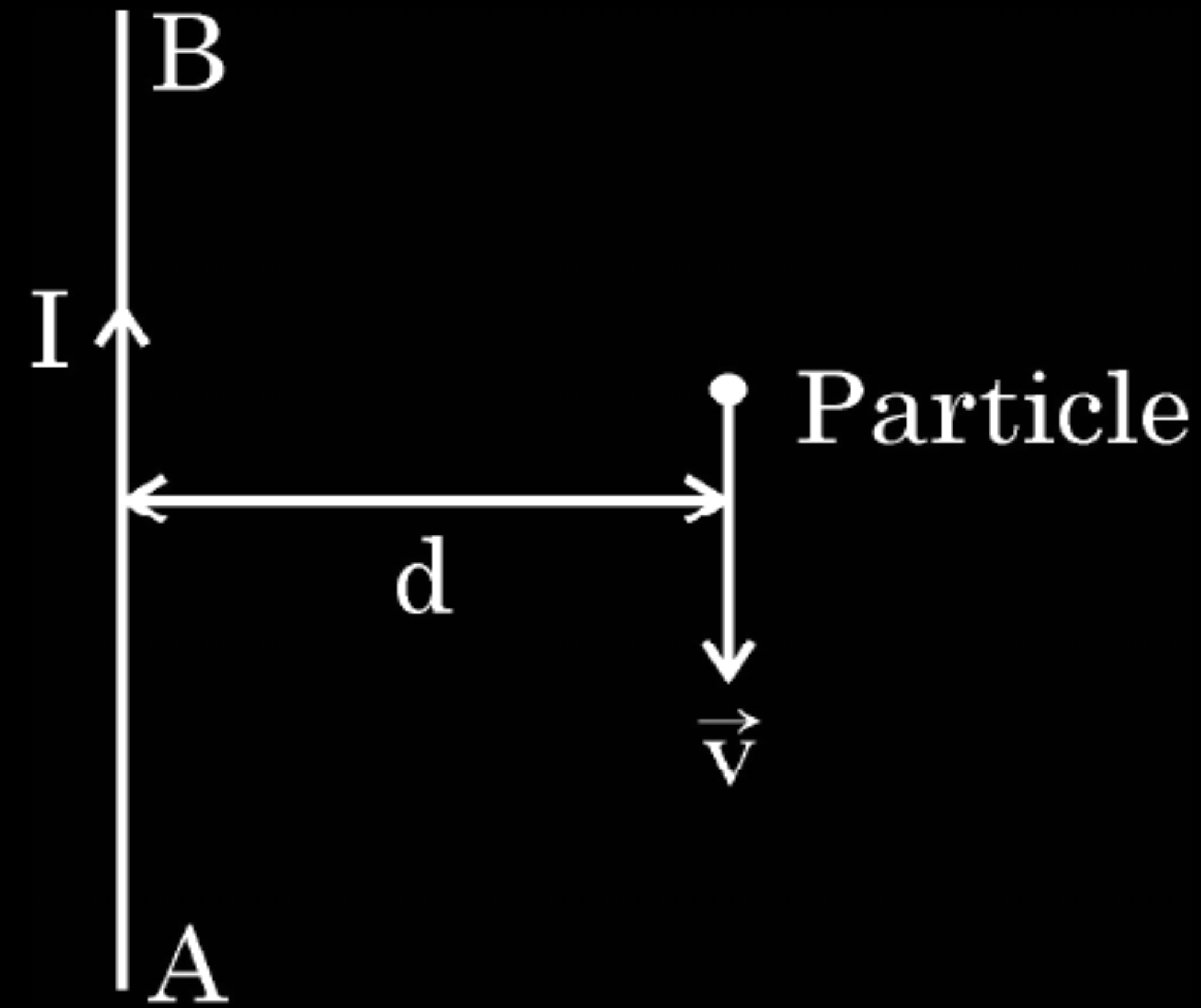
(C) $9 \times 10^{-7} \text{ N/m}$, repulsive

(D) $9 \times 10^{-5} \text{ N/m}$, attractive

23. (a) Write the expression for the Lorentz force on a particle of charge q moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$. When is the magnitude of this force maximum ? Show that no work is done by this force on the particle during its motion from a point $\vec{r}_1$ to point $\vec{r}_2$.

- (b) A long straight wire AB carries a current I . A particle (mass m and charge q) moves with a velocity $\vec{v}$, parallel to the wire, at a distance d from it as shown in the figure. Obtain the expression for the force experienced by the particle and mention its directions.

2



29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.
- (b) A galvanometer shows full scale deflection for current I_g . A resistance R_1 is required to convert it into a voltmeter of range $(0 - V)$ and a resistance R_2 to convert it into a voltmeter of range $(0 - 2V)$. Find the resistance of the galvanometer.

2. Two long parallel wires kept 2 m apart carry 3A current each, in the same direction. The force per unit length on one wire due to the other is **1**
- (A) $4.5 \times 10^{-5} \text{ Nm}^{-1}$, attractive (B) $4.5 \times 10^{-7} \text{ N/m}$, repulsive
- (C) $9 \times 10^{-7} \text{ N/m}$, repulsive (D) $9 \times 10^{-5} \text{ N/m}$, attractive

5. Two long straight parallel conductors carrying steady currents I_1 and I_2 in the same direction are separated by a distance d . The force exerted by one conductor on unit length of the other is :

(a) $\frac{\mu_0 I_1 I_2}{4\pi d}$, attractive

(b) $\frac{\mu_0 I_1 I_2}{4\pi d}$, repulsive

(c) $\frac{\mu_0 I_1 I_2}{2\pi d}$, attractive

(d) $\frac{\mu_0 I_1 I_2}{2\pi d}$, repulsive

- 6.** A long solenoid of length L and number of turns N carrying a current I produces a magnetic field B along its axis. If the current in the solenoid is doubled and its number of turns halved, the magnetic field along its axis will be :

(a) $2B$

(b) $4B$

(c) $\frac{B}{2}$

(d) B

- 32.** (a) (i) A charged particle with speed v_0 enters a region in which a strong and non-uniform magnetic field exists everywhere. It comes out of the region following a complicated trajectory without suffering any collision in the region. Would its final speed v be equal to its initial speed v_0 ? Justify your answer.
- (ii) In a region, a uniform magnetic field of $6 \times 10^{-4} \text{ T}$ is maintained. An electron enters the field with a speed of $3 \times 10^6 \text{ ms}^{-1}$ normal to the field. Find radius of its path. Also calculate its energy in eV.

- (b) (i) Derive an expression for the force acting on a conductor carrying current I in a magnetic field $\vec{B}$.
- (ii) A long horizontal conductor carries a current of 30 A, in east to west direction. What are the magnitude and direction of the magnetic field due to the current 1.0 m below the conductor ?

17. *Assertion (A) :* A proton and an alpha particle having same kinetic energy are moving in circular paths in a uniform magnetic field. The radii of their circular paths will be equal.

Reason (R) : The centripetal force required to move a charged particle in a circle does not depend on the magnitude of the magnetic field.

- 29.** (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?
- (b) Why should the value of resistance connected in parallel to a galvanometer be low ?
- (c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ?

2. An electron with velocity $\vec{v} = (v_x \hat{i} + v_y \hat{j})$ moves through a magnetic field $\vec{B} = (B_x \hat{i} - B_y \hat{j})$. The force $\vec{F}$ on the electron is : (e is the magnitude of its charge)

(a) $-e (v_x B_y - v_y B_x) \hat{k}$

(b) $e (v_x B_y - v_y B_x) \hat{k}$

(c) $-e (v_x B_y + v_y B_x) \hat{k}$

(d) $e (v_x B_y + v_y B_x) \hat{k}$

17. *Assertion (A) :* The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason (R): The magnetic field inside the solenoid is uniform.

17. *Assertion (A) :* A planar loop of irregular shape carrying current is subjected to a magnetic field acting perpendicular to the plane of the loop. If the wire is flexible, the loop takes a circular shape.

Reason (R) : The force acting on each point of a current carrying loop, in a magnetic field perpendicular to its plane, is radially outward.

2. Two insulated concentric coils, each of radius R , placed at right angles to each other, carry currents I and $\sqrt{3} I$ respectively. The magnitude of the net magnetic field at their common centre will be

(A) $\frac{\mu_0 I}{R}$

(B) $\frac{\mu_0 I}{2R}$

(C) $\frac{\mu_0 I}{4R}$

(D) $\frac{2\mu_0 I}{R}$

4. A galvanometer of resistance $100\ \Omega$ gives full scale deflection for a current of $1.0\ \text{mA}$. It is converted into an ammeter of range $(0 - 1\text{A})$. The resistance of the ammeter will be close to

(A) $0.1\ \Omega$

(B) $0.8\ \Omega$

(C) $1.0\ \Omega$

(D) $10\ \Omega$

16. **Assertion (A)** : The energy of a charged particle moving in a magnetic field does not change.

Reason (R) : It is because the work done by the magnetic force on the charge moving in a magnetic field is zero.

17. Derive an expression for magnetic force $\vec{F}$ acting on a straight conductor of length L carrying current I in an external magnetic field $\vec{B}$. Is it valid when the conductor is in zig-zag form ? Justify.

22. (b) (i) State and explain Ampere's circuital law.
- (ii) Two long straight parallel wires separated by 20 cm, carry 5 A and 10 A current respectively, in the same direction. Find the magnitude and direction of the net magnetic field at a point midway between them.

27. An electron is moving with a velocity $\vec{v} = \left(3 \times 10^6 \frac{\text{m}}{\text{s}}\right)\hat{i}$. It enters a region of magnetic field $\vec{B} = (91 \text{ mT})\hat{k}$.

- (a) Calculate the magnetic force $\vec{F}_B$ acting on electron and the radius of its path.
- (b) Trace the path described by it.

1. An ammeter and a voltmeter are connected in series to a battery. Their readings are noted as 'A' and 'V' respectively. If a resistor is connected in parallel with the voltmeter, then
 - (A) A will increase, V will decrease.
 - (B) A will decrease, V will increase.
 - (C) Both A and V will decrease.
 - (D) Both A and V will increase.

4. A galvanometer shows full scale deflection for a current I_g . If a shunt of resistance S_1 is connected to the galvanometer, it gets converted into an ammeter of range $(0 - I)$. When resistance of the shunt is made S_2 , its range becomes $(0 - 2I)$. Then $\left(\frac{S_1}{S_2}\right)$ is

(A) $\frac{I + I_g}{I - I_g}$

(B) $\frac{I - I_g}{I + I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

(D) $\frac{I - I_g}{2I - I_g}$

2. A particle of mass m and charge q describes a circular path of radius R in a magnetic field. If its mass and charge were $2m$ and $\frac{q}{2}$ respectively, the radius of its path would be

(A) $\frac{R}{4}$

(B) $\frac{R}{2}$

(C) $2R$

(D) $4R$

4. A galvanometer of resistance $50\ \Omega$ is converted into a voltmeter of range $(0 - 2\text{V})$ using a resistor of $1.0\ \text{k}\Omega$. If it is to be converted into a voltmeter of range $(0 - 10\ \text{V})$, the resistance required will be
- | | |
|---------------------------|---------------------------|
| (A) $4.8\ \text{k}\Omega$ | (B) $5.0\ \text{k}\Omega$ |
| (C) $5.2\ \text{k}\Omega$ | (D) $5.4\ \text{k}\Omega$ |

5. Two long straight parallel conductors A and B, kept at a distance r , carry current I in opposite directions. A third identical conductor C, kept at a distance $\left(\frac{r}{3}\right)$ from A carry current I_1 in the same direction as in A. The net magnetic force on unit length of C is

1

(A) $\frac{3\mu_0 I I_1}{2\pi r}$, towards A

(B) $\frac{3\mu_0 I I_1}{2\pi r}$, towards B

(C) $\frac{3\mu_0 I I_1}{4\pi r}$, towards A

(D) $\frac{3\mu_0 I I_1}{4\pi r}$, towards B

8. A galvanometer of resistance $100\ \Omega$ is converted into an ammeter of range $(0 - 1\ \text{A})$ using a resistance of $0.1\ \Omega$. The ammeter will show full scale deflection for a current of about

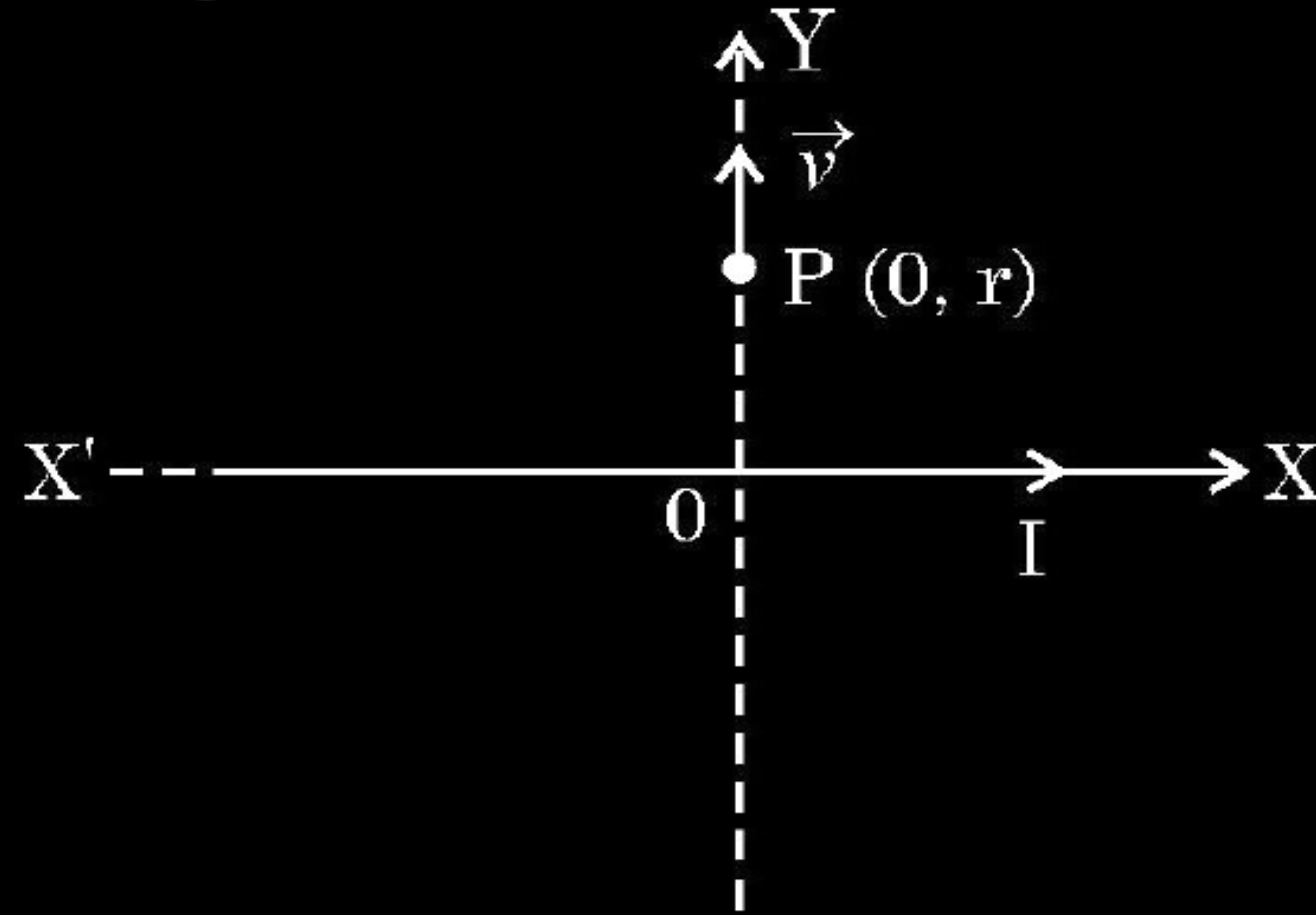
(A) $0.1\ \text{mA}$

(B) $1\ \text{mA}$

(C) $10\ \text{mA}$

(D) $0.1\ \text{A}$

26. An infinite straight conductor is kept along $X'X$ axis and carries a current I . A charge q at point $P(0, r)$ starts moving with velocity $\vec{v} = v_0 \hat{j}$ as shown in figure. Find the direction and magnitude of force initially experienced by the charge.



- (ii) An electron is revolving around a proton in an orbit of radius r with a speed v . Obtain expression for magnetic moment associated with the electron.

5. A particle of mass m and charge q is moving with velocity $\vec{v} = v_x \hat{i} + v_y \hat{j}$.

If it is subjected to a magnetic field $\vec{B} = B_0 \hat{i}$, it will move in a –

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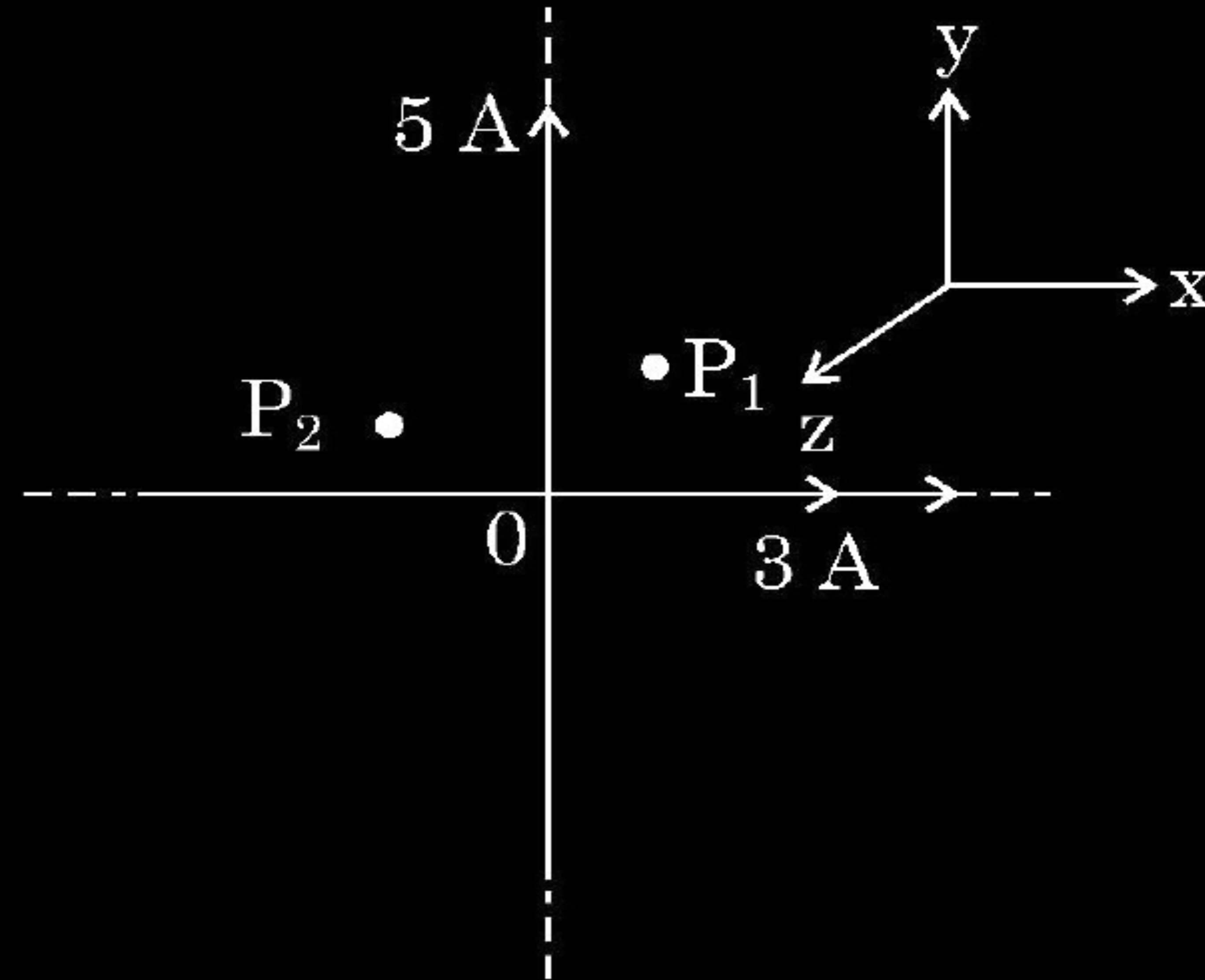
(A) straight line path

(B) circular path

(C) helical path

(D) parabolic path

25. Two long insulated straight wires carrying currents of 3 A and 5 A are arranged in XY plane as shown in figure. Find the magnitude and direction of the net magnetic fields at points $P_1(2\text{m}, 2\text{m})$ and $P_2(-1\text{m}, 1\text{m})$. 3



5. A straight wire is kept horizontally along east-west direction. If a steady current flows in wire from east to west, the magnetic field at a point above the wire will point towards

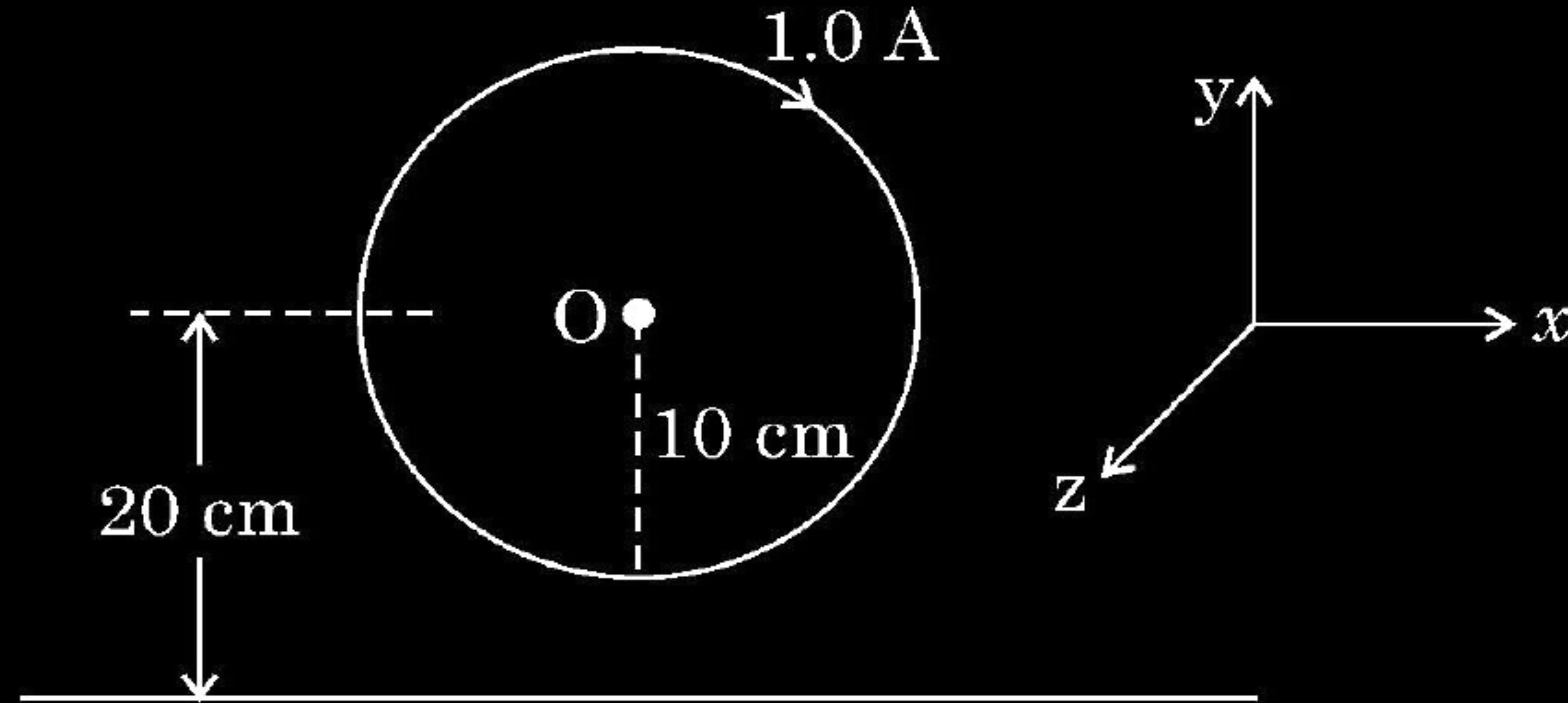
(A) East

(B) West

(C) North

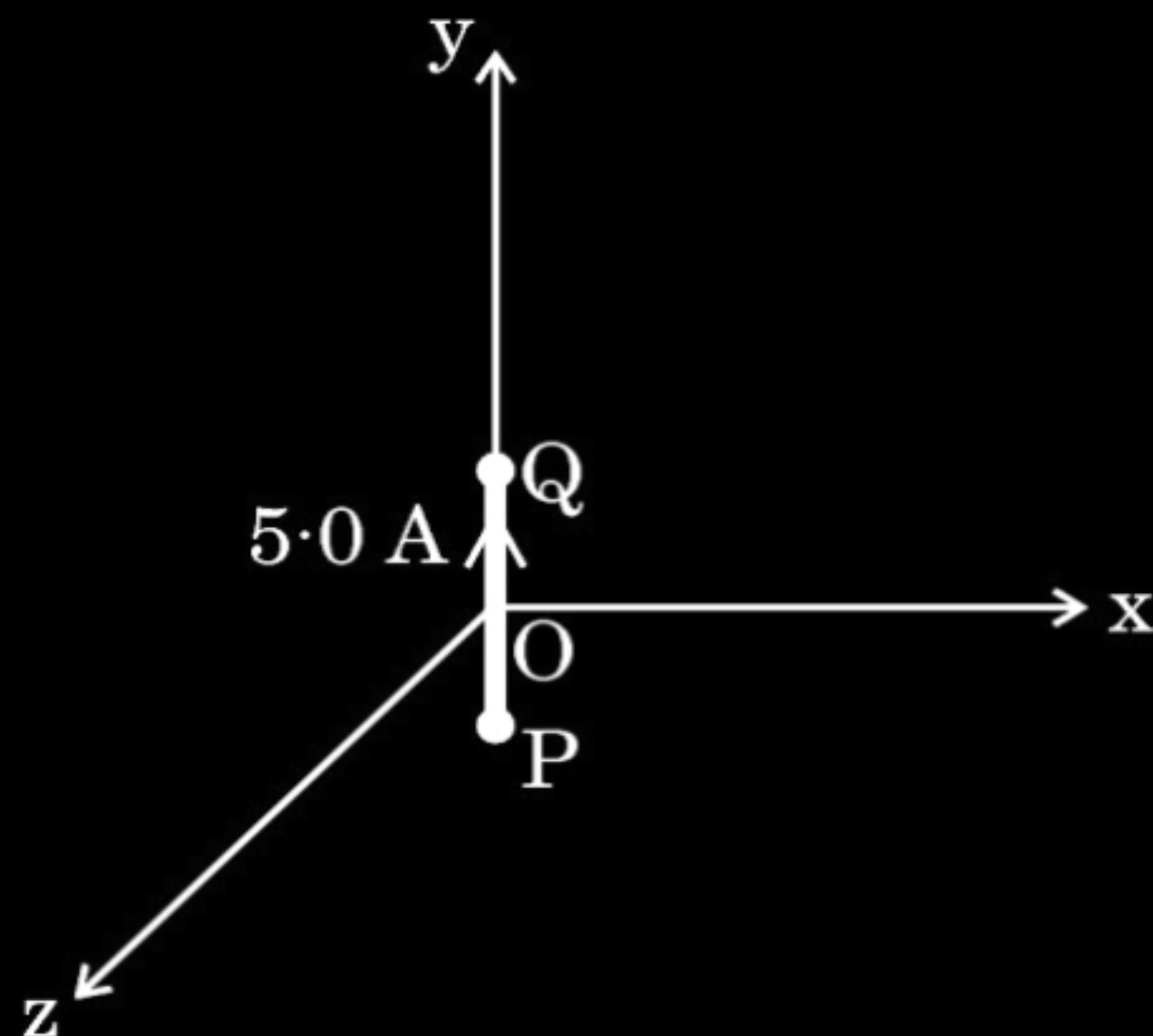
(D) South

26. A circular loop of radius 10 cm carrying current of 1.0 A lies in x - y plane. A long straight wire lies in the same plane parallel to x -axis at a distance of 20 cm as shown in figure.



Find the direction and value of current that has to be maintained in the wire so that the net magnetic field at O is zero.

3. A 2.0 cm segment of wire, carrying 5.0 A current in positive y-direction lies along y-axis, as shown in the figure. The magnetic field at a point (3 m, 4 m, 0) due to this segment (part of a circuit) is :



- (A) $(0.12 \text{ nT}) \hat{j}$
(B) $-(0.10 \text{ nT}) \hat{j}$
(C) $-(0.24 \text{ nT}) \hat{k}$
(D) $(0.24 \text{ nT}) \hat{k}$

- (a) (i) (1) What is meant by current sensitivity of a galvanometer ?
Mention the factors on which it depends.
- (2) A galvanometer of resistance G is converted into a voltmeter of range $(0 - V)$ by using a resistance R . Find the resistance, in terms of R and G , required to convert it into a voltmeter of range $\left(0 - \frac{V}{2}\right)$.

4. Two thin long parallel wires separated by a distance 'a' carry current 'I' in opposite directions. The wires will :

- (A) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (B) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (C) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.
- (D) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

15. *Assertion (A) :* A proton and an electron enter a uniform magnetic field $\vec{B}$ with the same momentum $\vec{p}$ such that $\vec{p}$ is perpendicular to $\vec{B}$. They describe circular paths of the same radius.

Reason (R) : In a magnetic field, orbital radius r is equal to $\frac{p}{qB}$.

- 23.** A rectangular loop of area $\vec{A}$, carrying current I , is placed in a uniform magnetic field $\vec{B}$. With the help of a suitable diagram, derive an expression, in vector form, for the torque acting on the loop.

4. A circular loop of wire, carrying a current 'I' is lying in xy-plane with its centre coinciding with the origin. It is subjected to a uniform magnetic field pointing along + z-axis. The loop will :
- | | |
|-----------------------|-------------------------|
| (A) move along x-axis | (B) move along - y-axis |
| (C) move along z-axis | (D) remain stationary |

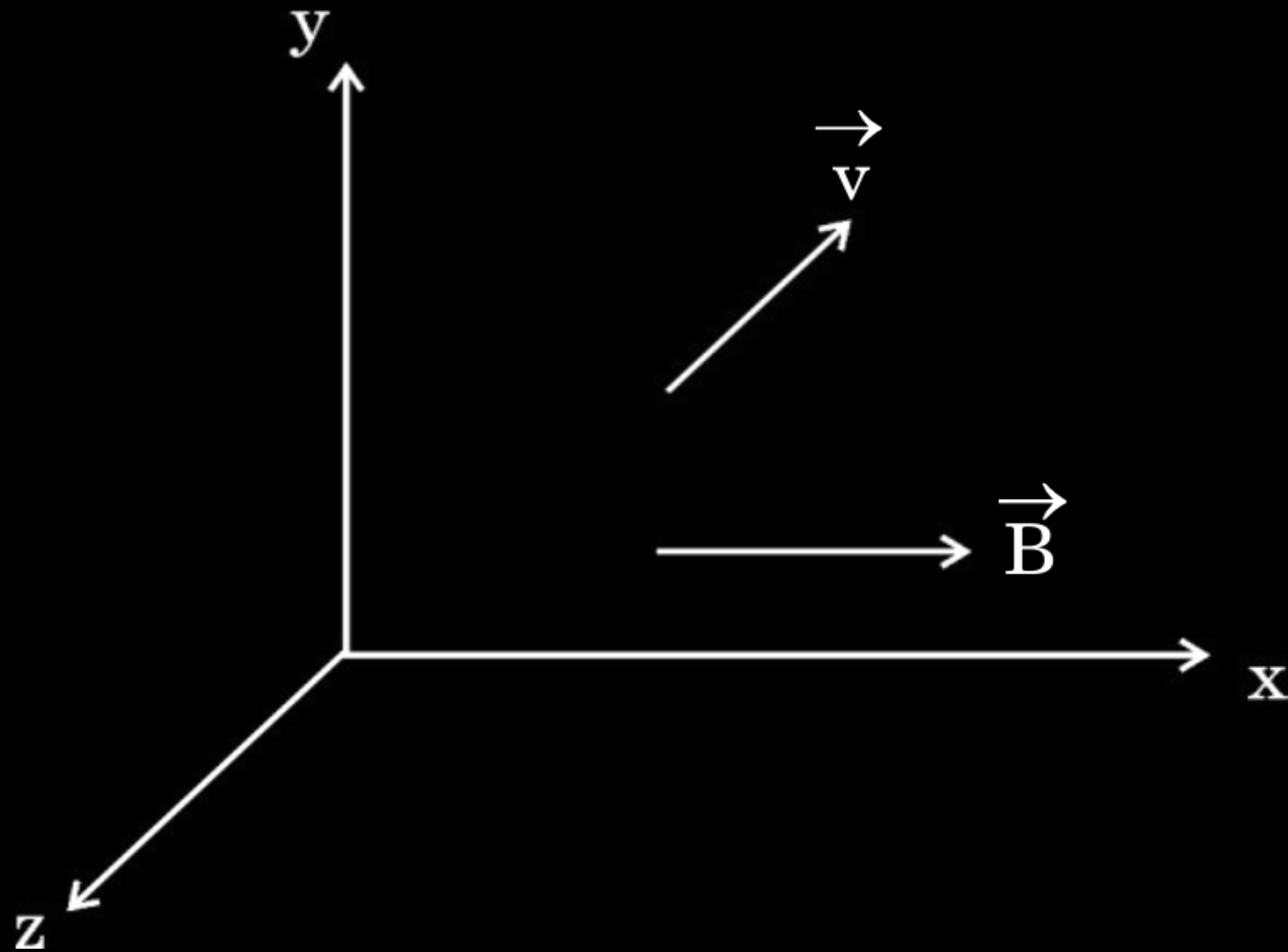
23. Two long straight parallel conductors carrying currents, exert a force on each other. Why ? Derive an expression for the force per unit length between two long straight parallel conductors carrying currents in opposite directions. Explain the nature of the force between these conductors.

- 22.** (a) Two long, straight, parallel conductors carry steady currents in opposite directions. Explain the nature of the force of interaction between them. Obtain an expression for the magnitude of the force between the two conductors. Hence define one ampere. 3

OR

- (b) Obtain an expression for the torque $\vec{\tau}$ acting on a current carrying loop in a uniform magnetic field $\vec{B}$. Draw the necessary diagram. 3

- 33.** (a) (i) A particle of mass m and charge q is moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$ as shown in the figure. Show that it follows a helical path. Hence, obtain its frequency of revolution.



- (ii) In a hydrogen atom, the electron moves in an orbit of radius $2\text{ \AA}$ making 8×10^{14} revolutions per second. Find the magnetic moment associated with the orbital motion of the electron.

- (b) (i) What is current sensitivity of a galvanometer ? Show how the current sensitivity of a galvanometer may be increased. “Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity.” Explain.
- (ii) A moving coil galvanometer has a resistance $15\ \Omega$ and takes $20\ \text{mA}$ to produce full scale deflection. How can this galvanometer be converted into a voltmeter of range 0 to $100\ \text{V}$?

14. *Assertion (A) :* An electron and a proton enter with the same momentum $\vec{p}$ in a magnetic field $\vec{B}$ such that $\vec{p} \perp \vec{B}$. Then both describe a circular path of the same radius.

Reason (R) : The radius of the circular path described by the charged particle (charge q , mass m) moving in the magnetic field $\vec{B}$ is given by $r = \frac{mv}{qB}$.

3. A wire of length 4.4 m is bent round in the shape of a circular loop and carries a current of 1.0 A. The magnetic moment of the loop will be :

(A) 0.7 Am^2

(B) 1.54 Am^2

(C) 2.10 Am^2

(D) 3.5 Am^2

4. Two charged particles, P and Q, each having charge q but of masses m_1 and m_2 , are accelerated through the same potential difference V . They enter a region of magnetic field $\vec{B} (\perp \vec{v})$ and describe the circular paths of radii a and b respectively. Then $\left(\frac{m_1}{m_2}\right)$ is equal to :

(A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\left(\frac{a}{b}\right)^2$ (D) $\left(\frac{b}{a}\right)^2$

5. A galvanometer of resistance $G \Omega$ is converted into an ammeter of range 0 to I A. If the current through the galvanometer is 0.1% of I A, the resistance of the ammeter is :

(A) $\frac{G}{999} \Omega$ (B) $\frac{G}{1000} \Omega$ (C) $\frac{G}{1001} \Omega$ (D) $\frac{G}{100.1} \Omega$

6. A 10 cm long wire lies along y -axis. It carries a current of 1.0 A in positive y -direction. A magnetic field $\vec{B} = (5 \text{ mT})\hat{j} - (8 \text{ mT})\hat{k}$ exists in the region. The force on the wire is :

(A) $(0.8 \text{ mN})\hat{i}$

(B) $-(0.8 \text{ mN})\hat{i}$

(C) $(80 \text{ mN})\hat{i}$

(D) $-(80 \text{ mN})\hat{i}$

13. *Assertion (A) :* Two long parallel wires, freely suspended and connected in series to a battery, move apart.

Reason (R) : Two wires carrying current in opposite directions repel each other.

- 24.** A circular coil with cross-sectional area 0.2 cm^2 carries a current of 4 A. It is kept in a uniform magnetic field of magnitude 0.5 T normal to the plane of the coil. Calculate :
- (a) the net force on the coil.
 - (b) the torque on the coil.
 - (c) the average force on each electron in the coil due to the magnetic field. The free electron density in the material of the coil is 10^{28} m^{-3} .

4. A piece of wire bent in the form of a circular loop A carries a current I . The wire is then bent into a circular loop B of two turns and carries the same current. The ratio of magnetic fields at the centre of loop A to that of loop B will be :

(A) $\frac{1}{16}$

(B) 16

(C) 4

(D) $\frac{1}{4}$

4. A loop carrying a current I clockwise is placed in $x - y$ plane, in a uniform magnetic field directed along z -axis. The tendency of the loop will be to :
- (A) move along x -axis (B) move along y -axis
(C) shrink (D) expand

3. The magnetic field due to a current carrying circular coil at a point far off on its axis at a distance 'r' from the centre of the coil is proportional to :

(A) $\frac{1}{r}$

(B) $\frac{1}{r^2}$

(C) $\frac{1}{r^{3/2}}$

(D) $\frac{1}{r^3}$

4. The magnetic field B within the solenoid of length L having N turns and carrying current I will be :

(A) $\mu_0 N I L$ (B) $\frac{\mu_0 N I}{L}$

(C) $\frac{\mu_0 N I}{4\pi L}$ (D) $\mu_0 N I$

5. A circular loop of radius 'a' carries a current $3I$ through it. Another concentric circular loop of radius $2a$ carrying a current I is placed perpendicular to the first loop. The dipole moment of the equivalent magnetic dipole for the system is :

(A) $\pi I a^2$

(B) $5Ia$

(C) $5\pi I a^2$

(D) $7\pi I a^2$

15. *Assertion (A) :* When current is passed through a spring, two adjacent turns move towards each other.

Reason (R) : There is a force between any two adjacent turns of a current carrying spring.

- 23.** Two parallel straight wires of lengths 15 m and 20 m carrying currents 6.0 A and 8.0 A respectively in opposite directions, are 10 cm apart. Calculate the force on a 10 cm long section of the second wire near its centre. Also mention the nature of the force between these wires.

7. A current of 5 A is flowing in a wire of length 1.5 cm. A force of 7.5 mN acts on it when it is placed in a uniform magnetic field of 0.2 T. The angle between the magnetic field and the direction of current is :

(A) 30°

(B) 45°

(C) 60°

(D) 90°

16. *Assertion (A) :* The torque acting on a current carrying coil is maximum when it is suspended in a radial magnetic field.

Reason (R) : The torque tends to rotate the coil on its own axis.

- 18.** An electron is passing through a region and experiences no force. Under what condition is it possible when the region has (a) only the magnetic field, and (b) both the electric and the magnetic fields ? Justify your answers.

(a) An electron is revolving with speed v around the proton in a hydrogen atom, in a circular orbit of radius r . The magnitude of magnetic dipole moment of the electron is : I

(A) 4 evr

(B) 2 evr

(C) $\frac{1}{2}\text{ evr}$

(D) $\frac{1}{4}\text{ evr}$

(b) A square loop of side 5.0 cm carries a current of 2.0 A. The magnitude of magnetic dipole moment associated with the loop is : l

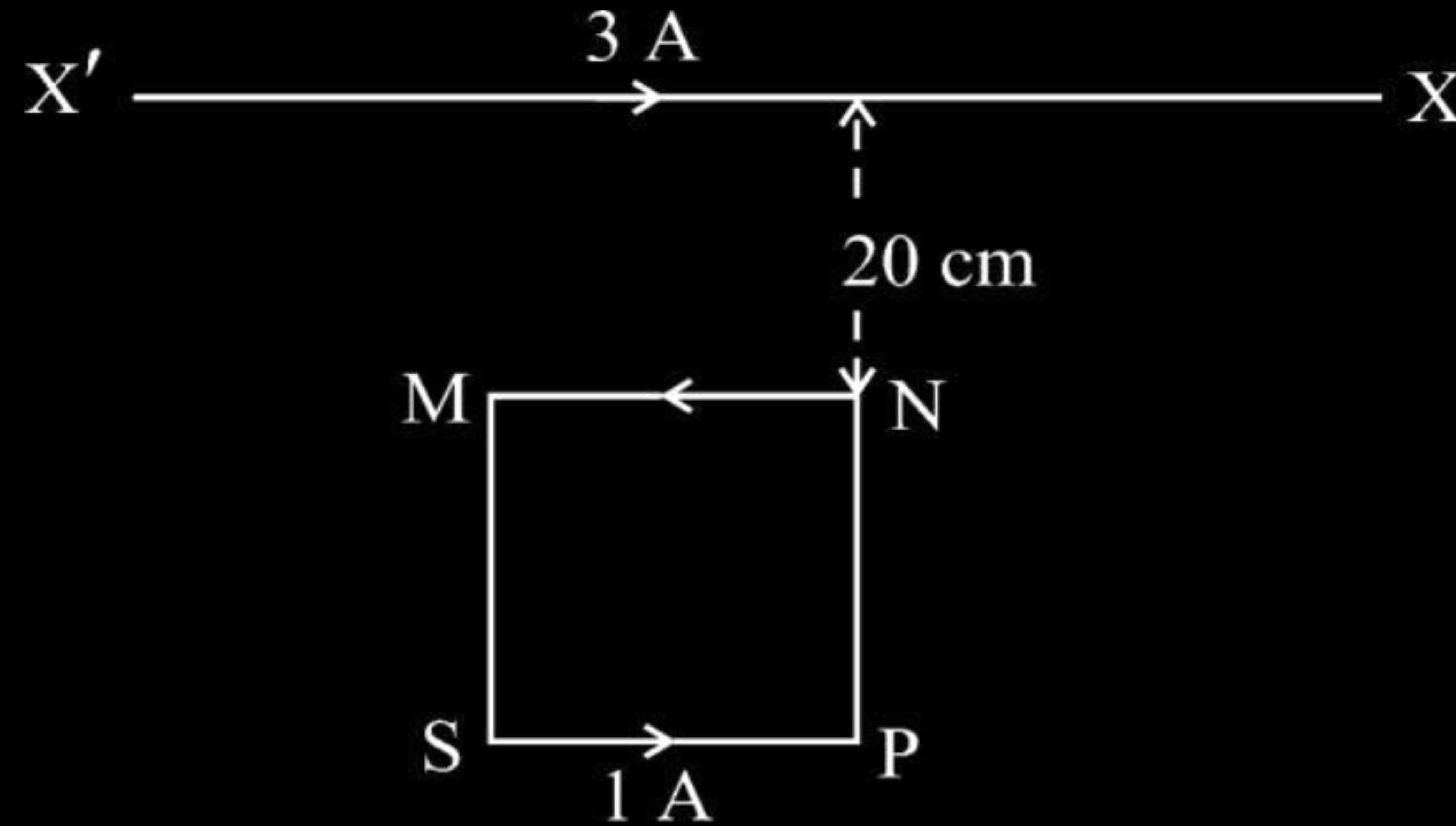
(A) $1.0 \times 10^{-3} \text{ Am}^2$

(B) $5.0 \times 10^{-3} \text{ Am}^2$

(C) $1.0 \times 10^{-2} \text{ Am}^2$

(D) $5.0 \times 10^{-2} \text{ Am}^2$

- (ii) A long, straight horizontal wire $X'X$ is held stationary and carries a current of 3.0 A . A square loop $MNPS$ of side 10 cm , carrying a current of 1.0 A is kept near the wire $X'X$ as shown in the figure. Find the magnitude and direction of the net magnetic force acting on the loop due to the wire.



2. Two circular loops of areas A and $4A$ carry currents $2I$ and I respectively. The magnetic fields at their centres will be in the ratio of :

(A) $3 : 1$

(B) $4 : 1$

(C) $1 : 1$

(D) $1 : 2$

12. The current sensitivity of a galvanometer does *not* depend on the :

- (A) magnetic field in which the coil is suspended.
- (B) current flowing in the coil.
- (C) torsional constant of the spring.
- (D) area of the coil.

22. A particle of mass m and charge q is moving in a magnetic field $\vec{B}$ with a velocity $\vec{v}$. Discuss, giving reasons, the shape of its trajectory when the angle between $\vec{v}$ and $\vec{B}$ is :

3

- (a) 0°
- (b) 90°
- (c) 120°

27. Two circular coils of radius R each and having equal number of turns N , carry equal currents I in the same direction. They are placed coaxially at a distance $2\sqrt{3} R$. Find the magnitude and direction of the net magnetic field produced at the midpoint of the line joining their centres.

7. A straight wire of length 1.0 m is placed along x-axis, in a region with magnetic field $\vec{B} = (3\hat{i} + 2\hat{j})$ T. A current of 2.0 A flows in the wire along +x direction. The magnetic force acting on the wire is :

(A) 2.0 N, along z-axis

(B) 2.0 N, along -z-axis

(C) 4.0 N, along z-axis

(D) 4.0 N, along -z-axis

- 18.** A long straight horizontal wire is carrying a current I . At an instant, an alpha particle at a distance r from it, is travelling with speed v parallel to the wire in a direction opposite to the current. Find the magnitude and direction of the force experienced by the particle at this instant.

5. Two parallel straight wires carrying currents I_1 and I_2 are separated by a distance x . The current in the first wire is increased from I_1 to $2I_1$ and their separation is reduced to $\frac{x}{2}$. The value of force between the wires will :
- (A) become four times
 - (B) become two times
 - (C) becomes one-fourth
 - (D) remain the same

15. *Assertion (A) :* A magnetic field does not interact with a moving charge.

Reason (R) : A moving charge does not produce a magnetic field.

32. (a) (i) Discuss the motion of a charged particle of mass m and charge q in a uniform magnetic field $\vec{B}$ with initial velocity $\vec{v}$
- (1) at an angle θ with field direction,
 - (2) perpendicular to the field direction ($\theta = 90^\circ$), and
 - (3) parallel to the field direction ($\theta = 0^\circ$).
- (ii) An electron is moving along a circular path of radius 18.2×10^{-2} m in a uniform magnetic field 2×10^{-3} T. Calculate :
- (1) the speed of the electron, and
 - (2) the potential difference through which the electron must be accelerated to acquire this speed.

- (b) (i) Deduce an expression for the torque experienced by a current carrying coil placed in an external uniform magnetic field.
- (ii) The maximum torque acting on a current carrying loop of area 0.08 m^2 is $8 \times 10^{-8} \text{ N-m}$. The current in the loop is $1.6 \mu\text{A}$. Calculate :
- (1) the magnetic field in which the loop is kept, and
 - (2) the magnetic dipole moment of the loop

14. *Assertion (A) :* A charged particle is moving with velocity v in x-y plane, making an angle θ ($0 < \theta < \frac{\pi}{2}$) with x-axis. If a uniform magnetic field $\vec{B}$ is applied in the region, along y-axis, the particle will move in a helical path with its axis parallel to x-axis.

Reason (R) : The direction of the magnetic force acting on a charged particle moving in a magnetic field is along the velocity of the particle.

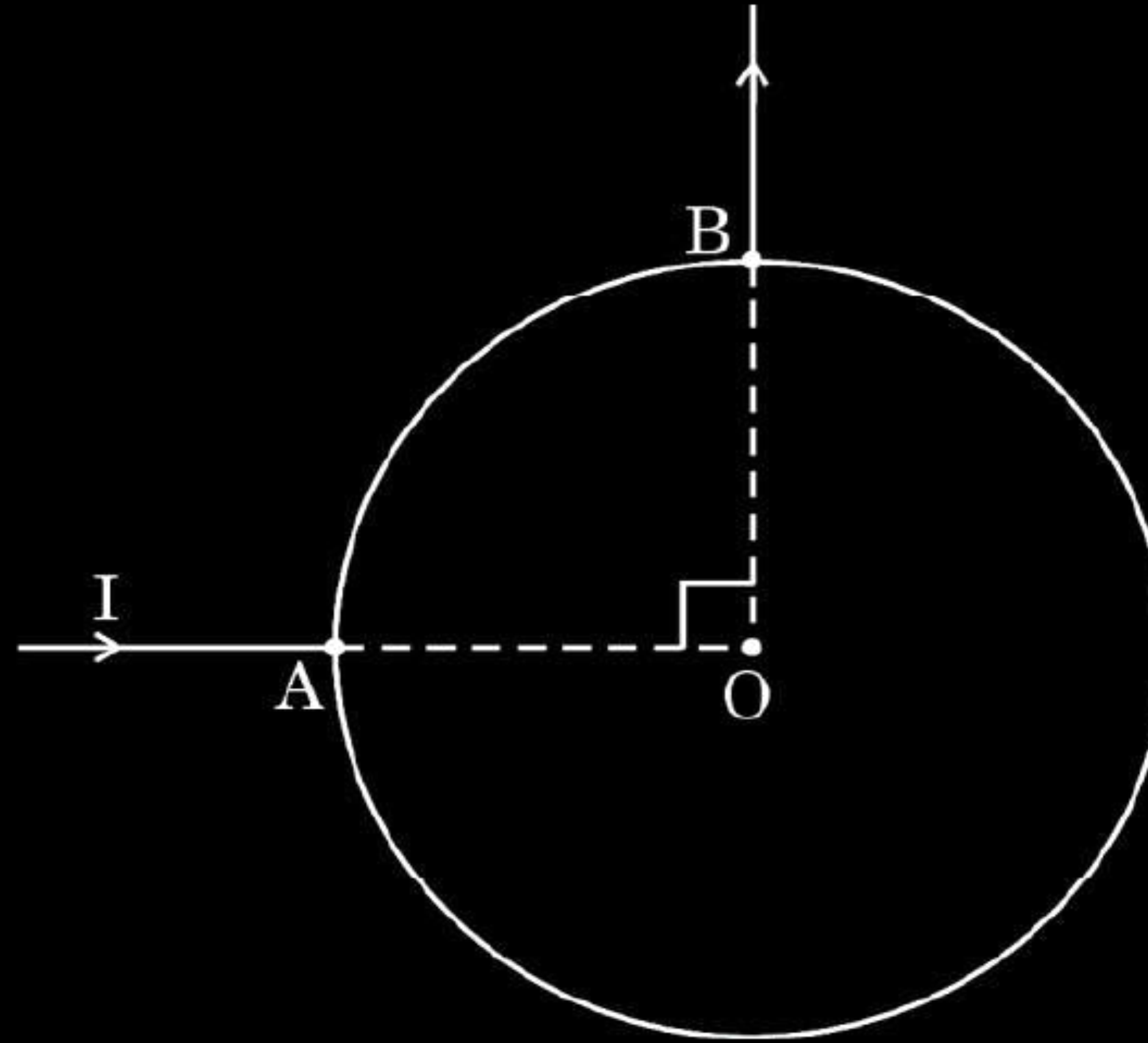
- 25.** (a) (i) Write Biot-Savart's law in vector form.
- (ii) Two identical circular coils A and B, each of radius R, carrying currents I and $\sqrt{3}I$ respectively, are placed concentrically in XY and YZ planes respectively. Find the magnitude and direction of the net magnetic field at their common centre.

- (b) (i) A rectangular loop of sides l and b carries a current I clockwise. Write the magnetic moment $\vec{m}$ of the loop and show its direction in a diagram.
- (ii) The loop is placed in a uniform magnetic field $\vec{B}$ and is free to rotate about an axis which is perpendicular to $\vec{B}$. Prove that the loop experiences no net force, but a torque $\vec{\tau} = \vec{m} \times \vec{B}$.

- 31.** (a) (i) With the help of a labelled diagram, explain the principle of working of a moving coil galvanometer. Write the purpose of using (i) radial magnetic field, and (ii) soft iron core, in it.
- (ii) Define current sensitivity of a galvanometer. “Increasing the current sensitivity may not necessarily increase the voltage sensitivity.” Give reason.

- (b) (i) (I) Write Ampere's circuital law in mathematical form and explain the terms used.
- (II) As the current carrying solenoid is made longer, the magnetic field produced outside it approaches zero. Why ?
- (III) A flexible loop of irregular shape carrying current when located in an external magnetic field, changes to a circular shape. Give reason.
- (ii) A galvanometer of resistance G is converted into a voltmeter to measure up to V volts, by connecting a resistance R_1 in series with the coil. If R_1 is replaced by R_2 , then it can only measure up to $\frac{V}{2}$ volt. Find the value of the resistance R_3 (in terms of R_1 and R_2) needed to convert it into a voltmeter that can read up to $2V$.

3. In a circular loop of radius R , current I enters at point A and exits at point B, as shown in the figure. The value of the magnetic field at the centre O of the loop is :



(A) $\frac{\mu_0 I}{R}$

(B) zero

(C) $\frac{\mu_0 I}{2R}$

(D) $\frac{\mu_0 I}{4R}$

7. A proton and an α -particle enter with the same velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ such that $\vec{v} \perp \vec{B}$. The ratio of the radii of their paths is :

(A) 2

(B) $\frac{1}{2}$

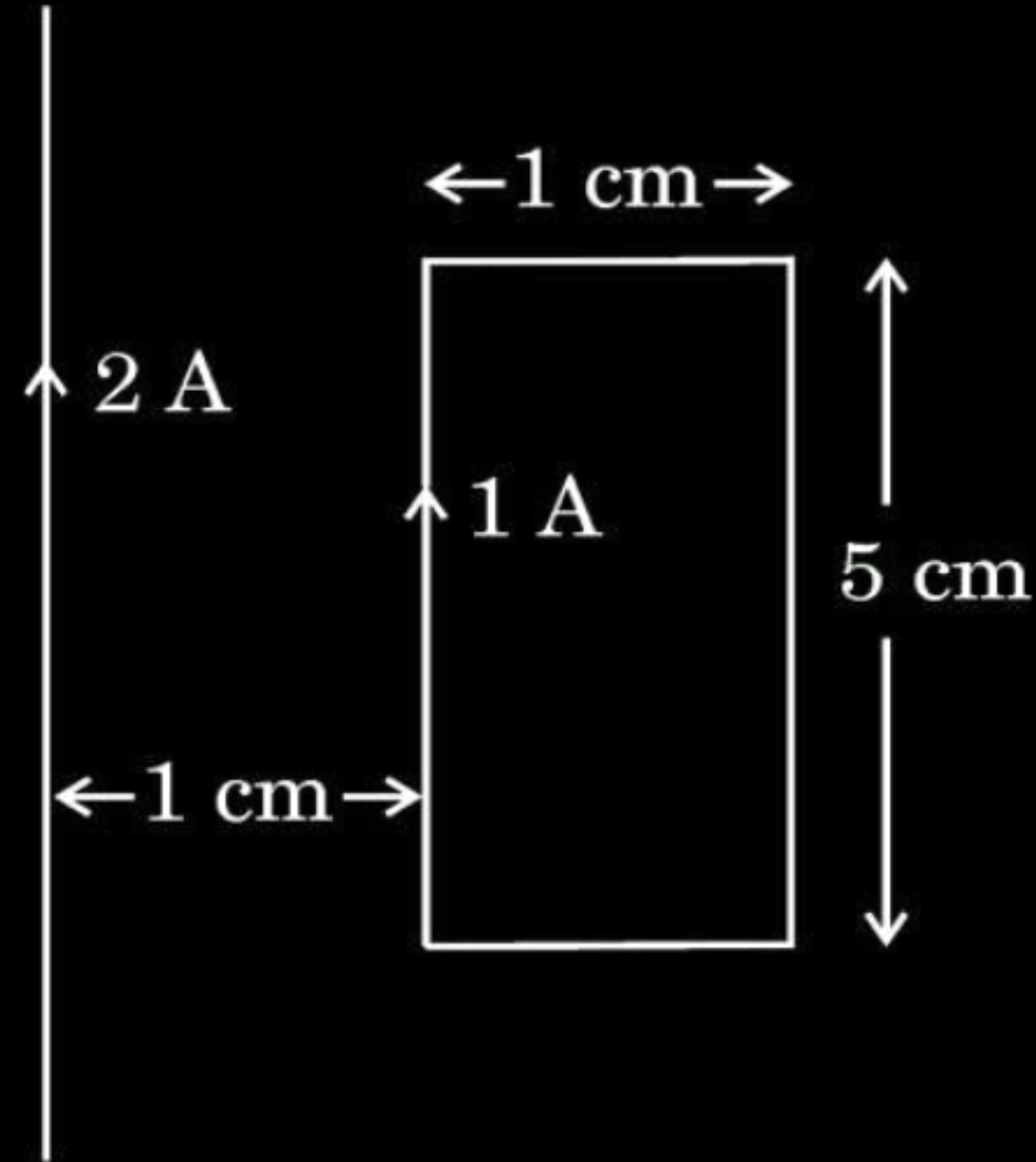
(C) $\frac{1}{4}$

(D) 4

11. A galvanometer can be converted into an ammeter of desired range by connecting a :

- | | |
|----------------------------------|----------------------------------|
| (A) small resistance in series | (B) large resistance in series |
| (C) small resistance in parallel | (D) large resistance in parallel |

24. A rectangular loop carries a current of 1 A. A straight long wire carrying 2 A current is kept near the loop in the same plane as shown in the figure.



Find :

- (i) the torque acting on the loop, and
- (ii) the magnitude and direction of the net force on the loop.

3. A straight conductor is carrying a current of 2 A in +x direction along it. A uniform magnetic field $\vec{B} = (0.6\hat{j} + 0.8\hat{k})$ T is switched on, in the region. The force acting on 10 cm length of the conductor is :

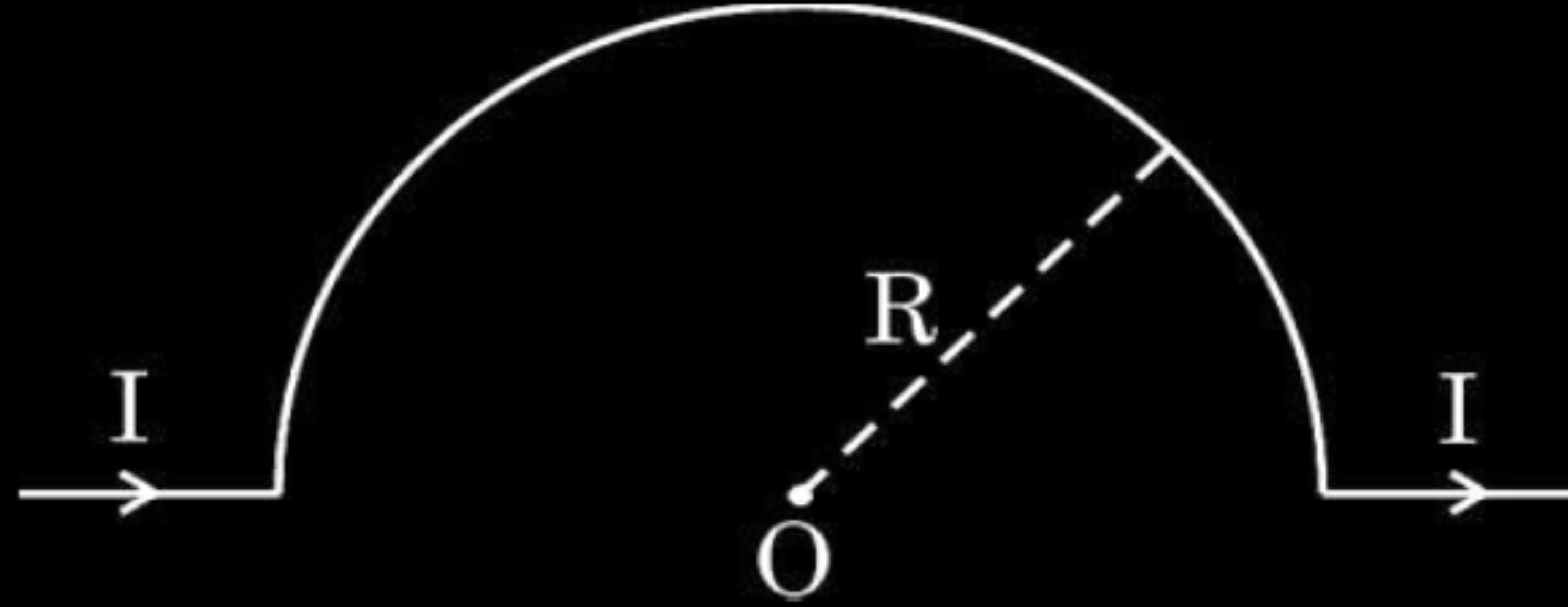
(A) $(0.12\hat{j} - 0.16\hat{k})$ N

(B) $(-0.16\hat{j} + 0.12\hat{k})$ N

(C) $(-0.12\hat{j} + 0.16\hat{k})$ N

(D) $(0.16\hat{j} - 0.12\hat{k})$ N

3. The value of magnetic field at point O in the given figure is :



(A) $\frac{\mu_0 I}{2\pi R}$

(B) $\frac{\mu_0 I}{\pi R}$

(C) $\frac{\mu_0 I}{4R}$

(D) $\frac{\mu_0 I}{R}$

3. A particle of mass m and charge q moving with a velocity $\vec{V}$ at right angle to a magnetic field $\vec{B}$, experiences a force $\vec{F}$. Another particle of mass $2m$ and charge $2q$ enters the same field $\vec{B}$ with a velocity $\frac{\vec{V}}{2}$, in the same direction. This particle will experience a force

(A) $4\vec{F}$

(B) $2\vec{F}$

(C) $\vec{F}$

(D) $\frac{\vec{F}}{2}$

4. A 5.0 cm long wire carrying 1.0 A current is kept in a region in which a uniform magnetic field of 0.3 T is applied perpendicular to the length of the wire. The magnitude of the force acting on the wire is –

1

(A) zero

(B) 0.45 N

(C) 0.03 N

(D) 0.015 N

23. (a) Write one point of similarity and one point of difference between magnetic field and the electrostatic field. **3**
- (b) A wire of length l is first bent into a circular loop and then into a circular coil of two turns. For the same current passing through them, find the ratio of the magnetic fields at their centres.

32. (a) (A) When a current passes through a conductor it produces a magnetic field.
- (i) State the right-hand thumb rule that gives the direction of the magnetic field.
 - (ii) How does the magnetic field vary with the distance from the conductor ?
 - (iii) Name the factors (other than the distance) which affect the magnitude of the magnetic field.

- (b) (A) A particle of mass m and charge q is moving with velocity v in a circular path under the influence of a uniform magnetic field $\vec{B}$. Explain how will the path followed be affected when
- (i) the strength of magnetic field is decreased.
 - (ii) the velocity of the particle is decreased.
 - (iii) the velocity of charged particle is such that one of its component is perpendicular to B and another component is parallel to $\vec{B}$.
 - (iv) an electric field is applied in such a way that electrostatic force balances the magnetic force on the charged particle.
 - (v) the magnetic field is suddenly removed.
- In each of the above cases, other factors remain the same.

(B) Derive an expression for torque acting as a rectangular loop carrying a steady current when placed in a uniform magnetic field.

6. A charged particle gains a speed of 10^6 ms^{-1} , when accelerated from rest through a potential difference 10 kV. It enters a region of magnetic field of 0.4 T such that $\vec{v} \perp \vec{B}$. The radius of circular path described by it is :
- (A) 2.5 cm (B) 5 cm
(C) 8 cm (D) 10 cm

7. A current of $\left(\frac{10}{\pi}\right)$ A is maintained in a circular loop of radius 14 cm. The value of dipole moment associated with the loop is :

(A) 0.019 Am^2

(B) 0.14 Am^2

(C) 0.196 Am^2

(D) 0.615 Am^2

18. A current of 5 A is passing along +X direction through a wire lying along X-axis. Find the magnetic field $\vec{B}$ at a point $\vec{r} = (3\hat{i} + 4\hat{j})$ m due to 1 cm element of the wire, centered at origin.

- 29.** A galvanometer is an instrument used to show the direction and strength of the current passing through it. In a galvanometer, a coil placed in a magnetic field experiences a torque and hence gets deflected when a current passes through it. The name is derived from the surname of Italian scientist L. Galvani, who in 1791 discovered that electric current makes a dead frog's leg jerk. A spring attached with the coil provides a counter torque.

In equilibrium, the deflecting torque is balanced by the restoring torque of the spring and we have :

$$NBAI = k\phi$$

where N is the total number of turns in the coil

A is the area of cross-section of each turn

B is the radial magnetic field

k is the torsional constant of the spring

ϕ is the angular deflection of the coil

As the current (I_g) which produces full scale deflection in the galvanometer is very small, the galvanometer cannot as such be used to measure current in electric circuits. A small resistance, called shunt, of a suitable value is connected with the galvanometer to convert it into an ammeter of desired range. By using a higher resistance, a galvanometer can also be converted into a voltmeter.

- (i) (a) A galvanometer is converted into a voltmeter of range $(0 - V)$ by connecting with it, a resistance R_1 . If R_1 is replaced by R_2 , the range becomes $(0 - 2 V)$. The resistance of the galvanometer is :

(A) $(R_2 - 2R_1)$

(B) $(R_2 - R_1)$

(C) $(R_1 + R_2)$

(D) $(R_1 - 2R_2)$

- (b) A current of 5 mA flows through a galvanometer. Its coil has 100 turns, each of area of cross-section 18 cm^2 and is suspended in a magnetic field 0.20 T. The deflecting torque acting on the coil will be :

1

- | | |
|-------------------------------------|-------------------------------------|
| (A) $3.6 \times 10^{-3} \text{ Nm}$ | (B) $1.8 \times 10^{-4} \text{ Nm}$ |
| (C) $2.4 \times 10^{-3} \text{ Nm}$ | (D) $1.2 \times 10^{-4} \text{ Nm}$ |

(ii) The value of resistance of the ammeter in case (ii) will be : *1*

(A) $0.20\ \Omega$

(B) $0.24\ \Omega$

(C) $6.0\ \Omega$

(D) $6.25\ \Omega$

(iii) A galvanometer of resistance $6\ \Omega$ shows full scale deflection for a current of 0.2 A . The value of shunt to be used with this galvanometer to convert it into an ammeter of range $(0 - 5\text{ A})$ is : *1*

(A) $0.25\ \Omega$

(B) $0.30\ \Omega$

(C) $0.50\ \Omega$

(D) $6.0\ \Omega$

(iv) The value of the current sensitivity of a galvanometer is given by : *1*

(A) $\frac{k}{NBA}$

(B) $\frac{NBA}{k}$

(C) $\frac{kBA}{N}$

(D) $\frac{kNB}{A}$

- 19.** A circular coil of wire having 200 turns, each of radius 4.0 cm is placed in a horizontal plane. It carries a current of 0.40 A in clockwise direction. Find the magnitude and direction of the magnetic field at the centre of the coil.

- 27.** (a) Use Ampere's law to derive the expression for the magnetic field due to a long straight current carrying wire of infinite length.
- (b) Why is Ampere's law used for the derivation in (a) above and not Biot-Savart's law ? Explain.

19. A voltmeter of resistance $1000\ \Omega$ can measure up to 25 V. How will you convert it so that it can read up to 250 V ?

26. An electron of mass m and charge $-e$ is revolving anticlockwise around the nucleus of an atom.

(a) Obtain the expression for the magnetic dipole moment (μ) of the atom.

(b) If $\vec{L}$ is the angular momentum of electron, show that

$$\vec{\mu} = - \left(\frac{e}{2m} \right) \vec{L} .$$

3. A particle of mass m and charge q moves along y -axis in a region in which a uniform magnetic field $\vec{B}$ is pointing along x -axis. The Lorentz force acting on the charge will point along :

(A) x -axis

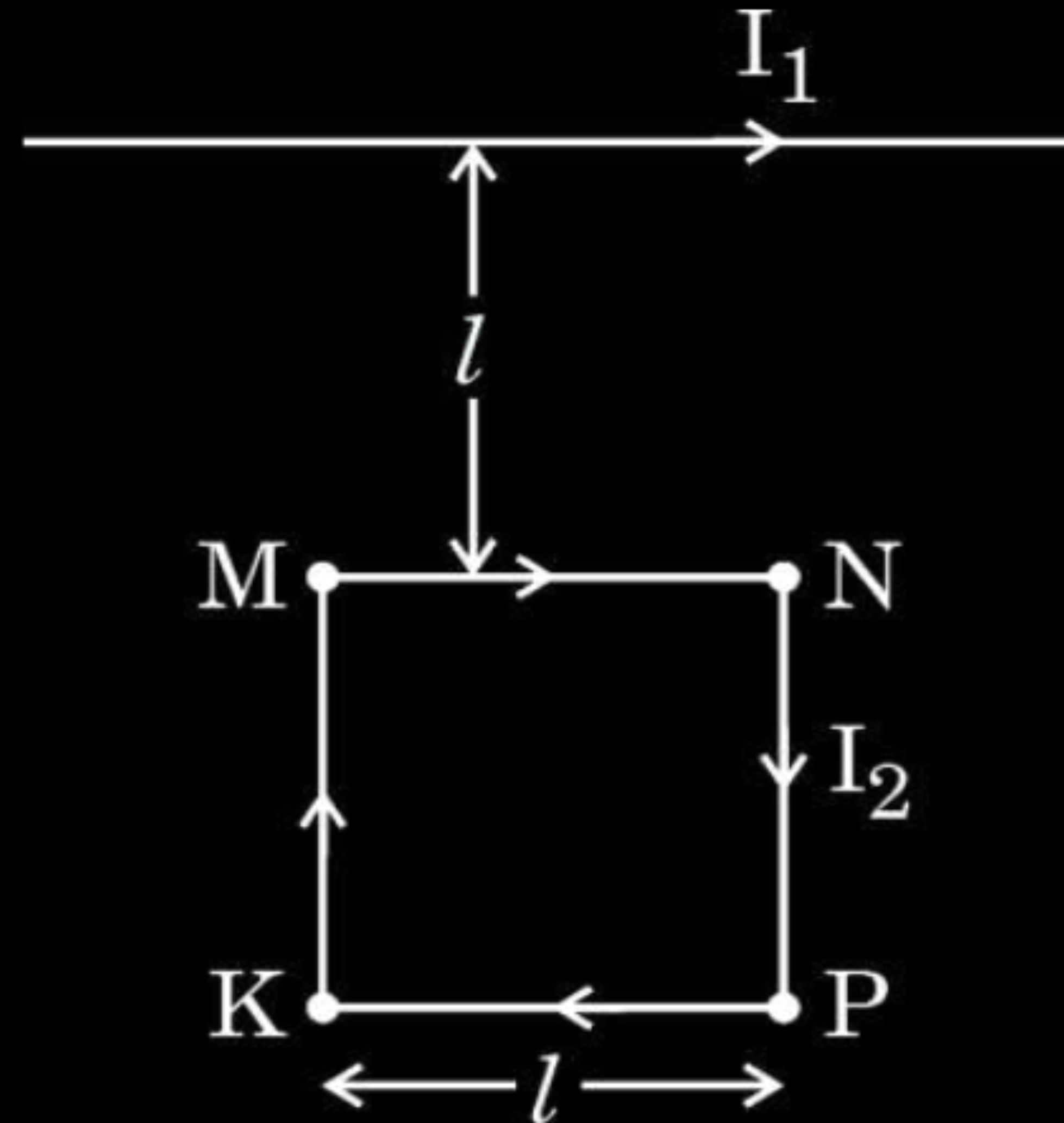
(B) y -axis

(C) z -axis

(D) negative z -axis

8. A long straight wire is held vertically and carries a steady current in upward direction. The shape of magnetic field lines produced by the current-carrying wire are :
- (A) horizontal straight lines directed radially out from the wire.
 - (B) straight lines parallel to the current-carrying wire.
 - (C) concentric horizontal circles around the wire.
 - (D) coaxial helixes around the wire.

23. (a) In which cases does a charged particle not experience a force in a magnetic field ?
- (b) A square loop MNPK of side ' l ' carrying a current ' I_2 ' is kept close to a long straight wire in the same plane and the wire carries a steady current I_1 as shown in the figure. Obtain the magnitude of magnetic force exerted by the wire on the loop.



3. A particle with charge q moving with velocity $\vec{v} = v_0 \hat{i}$ enters a region with magnetic field $\vec{B} = B_1 \hat{j} + B_2 \hat{k}$. The magnitude of force experienced by the particle is :

(A) $qv_0 (B_1 + B_2)$

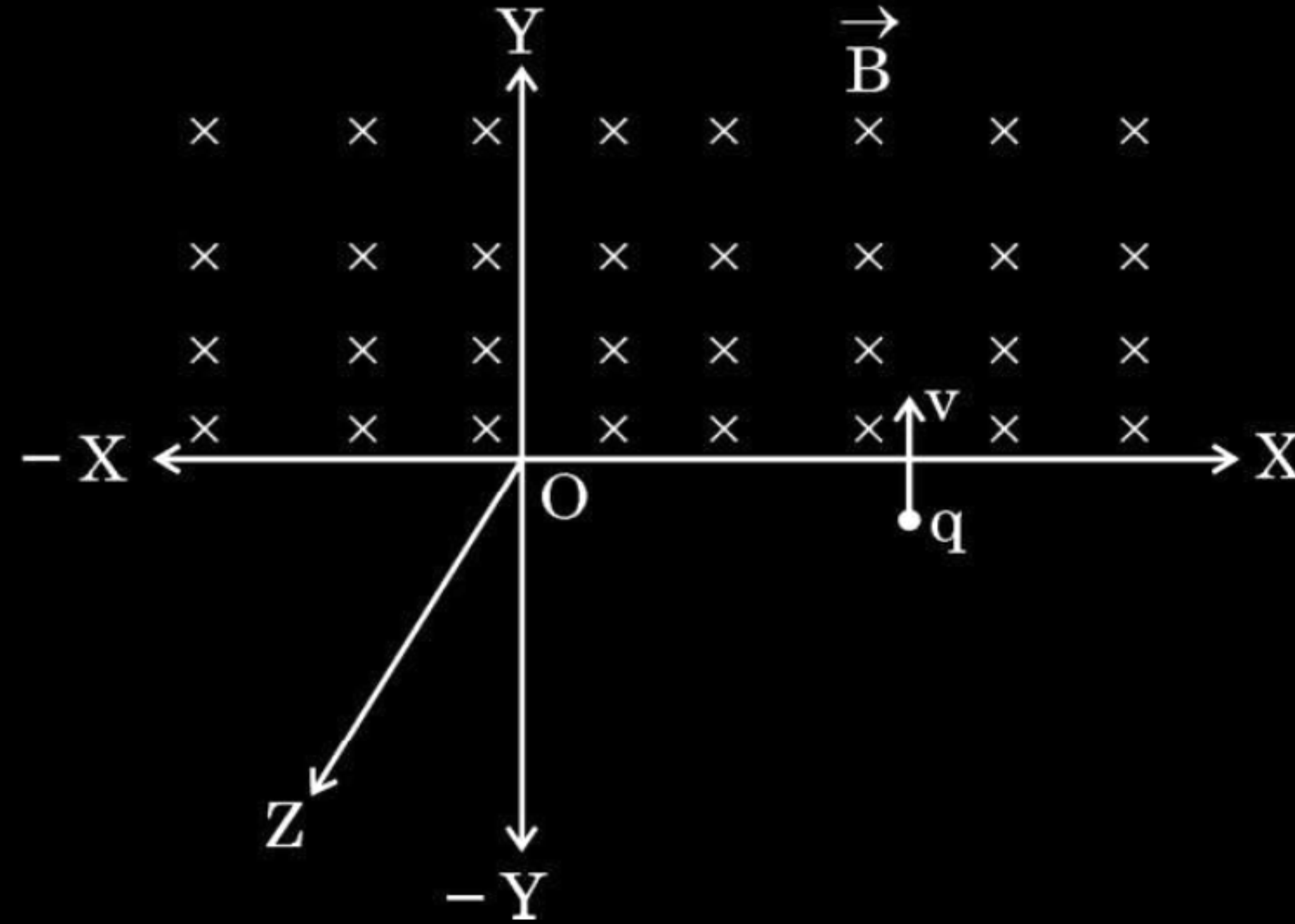
(B) $q\sqrt{v_0 (B_1 + B_2)}$

(C) $q v_0 \sqrt{(B_1^2 + B_2^2)}$

(D) $q \sqrt{v_0 (B_1^2 + B_2^2)}$

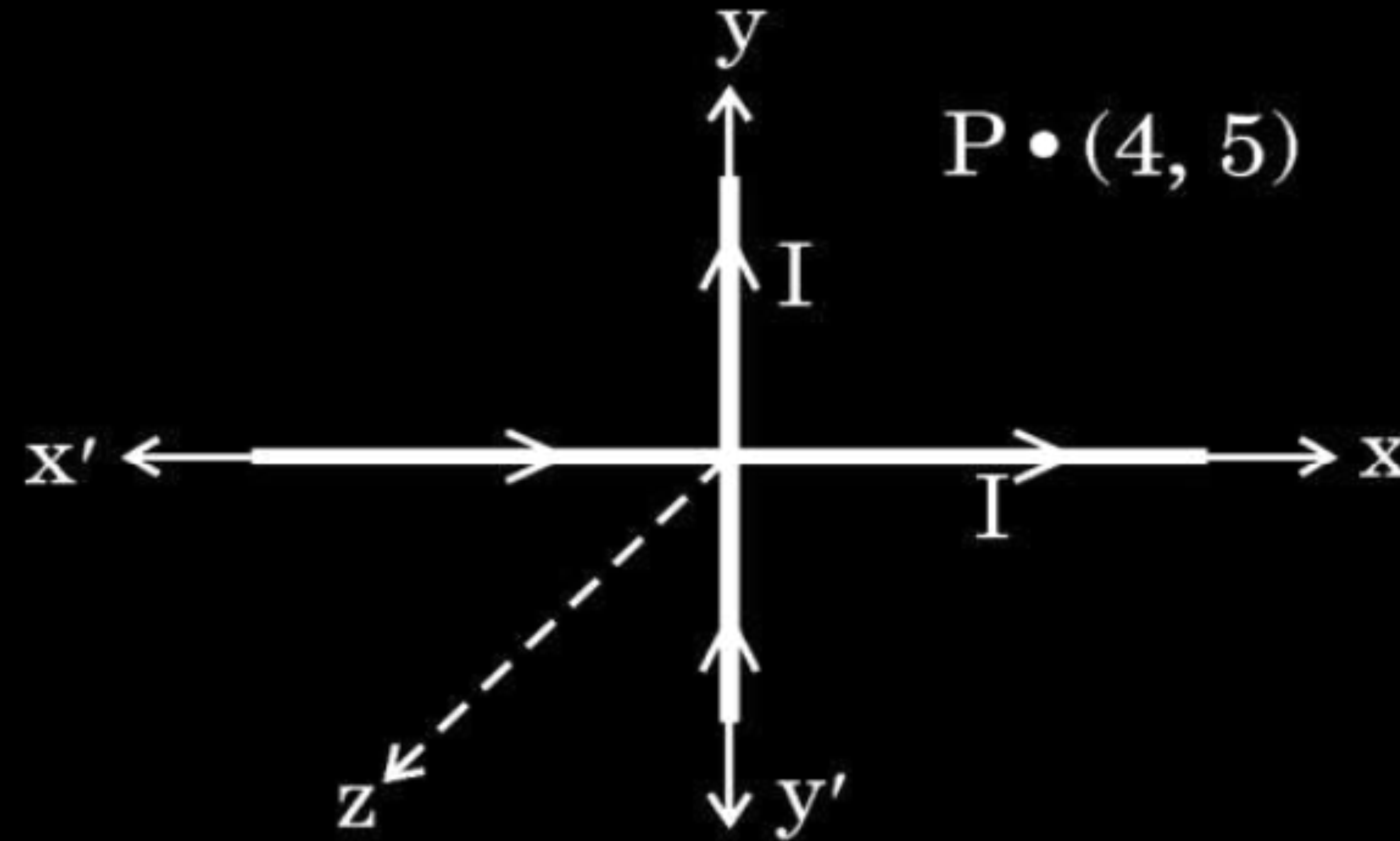
- 23.** (a) A charged particle q moving with a velocity $\vec{v}$ is subjected to a uniform magnetic field $\vec{B}$ acting perpendicular to $\vec{v}$. If a uniform electric field $\vec{E}$ is also set up in the region along the direction of $\vec{B}$, describe the path followed by the particle and draw its shape.
- (b) How will the magnetic field inside a long solenoid be affected when :
- (i) the radius of the turns of the solenoid is increased,
 - (ii) the length of solenoid as well as the total number of its turns are doubled ?

3. A particle having charge $+q$ enters a uniform magnetic field $\vec{B}$ as shown in the figure. The particle will describe :



- (A) a circular path in XZ plane
- (B) a semicircular path in XY plane
- (C) a helical path with its axis parallel to Y-axis
- (D) a semicircular path in YZ plane

- 23.** (a) Write vector form of Biot-Savart law.
- (b) Two insulated long straight wires, each carrying 2.0 A current are kept along xx' and yy' axis as shown in the figure. Find the magnitude and direction of resultant magnetic field at point P (4m, 5m).



6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is

1

(A) $1.0 \times 10^{-9} \hat{k} \text{ T}$

(B) $-1.0 \times 10^{-9} \hat{k} \text{ T}$

(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

(D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

27. (a) In a region of a uniform electric field $\vec{E}$, a negatively charged particle is moving with a constant velocity $\vec{v} = -v_0 \hat{i}$ near a long straight conductor coinciding with XX' axis and carrying current I towards $-X$ axis. The particle remains at a distance d from the conductor.
- (i) Draw diagram showing direction of electric and magnetic fields.
 - (ii) What are the various forces acting on the charged particle ?
 - (iii) Find the value of v_0 in terms of E , d and I .

- (b) Two infinitely long conductors kept along XX' and YY' axes are carrying current I_1 and I_2 along $-X$ axis and $-Y$ axis respectively. Find the magnitude and direction of the net magnetic field produced at point $P(X, Y)$.

31. (a) (i) What is the source of force acting on a current-carrying conductor placed in a magnetic field ? Obtain the expression for force acting between two long straight parallel conductors carrying steady currents and hence define 'ampere'.
- (ii) A point charge q is moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$. Find the work done by the magnetic force on the charge.
- (iii) Explain the necessary conditions in which the trajectory of a charged particle is helical in a uniform magnetic field.

- (b) (i) A current carrying loop can be considered as a magnetic dipole placed along its axis. Explain.
- (ii) Obtain the relation for magnetic dipole moment $\vec{M}$ of current carrying coil. Give the direction of $\vec{M}$.
- (iii) A current carrying coil is placed in an external uniform magnetic field. The coil is free to turn in the magnetic field. What is the net force acting on the coil ? Obtain the orientation of the coil in stable equilibrium. Show that in this orientation the flux of the total field (field produced by the loop + external field) through the coil is maximum.

6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is

1

(A) $1.0 \times 10^{-9} \hat{k} \text{ T}$

(B) $-1.0 \times 10^{-9} \hat{k} \text{ T}$

(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

(D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

3. A 1 cm segment of a wire lying along x-axis carries current of 0.5 A along +x direction. A magnetic field $\vec{B} = (0.4 \text{ mT}) \hat{j} + (0.6 \text{ mT}) \hat{k}$ is switched on, in the region. The force acting on the segment is

(A) $(2\hat{j} + 3\hat{k}) \text{ mN}$

(B) $(-3\hat{j} + 2\hat{k}) \mu\text{N}$

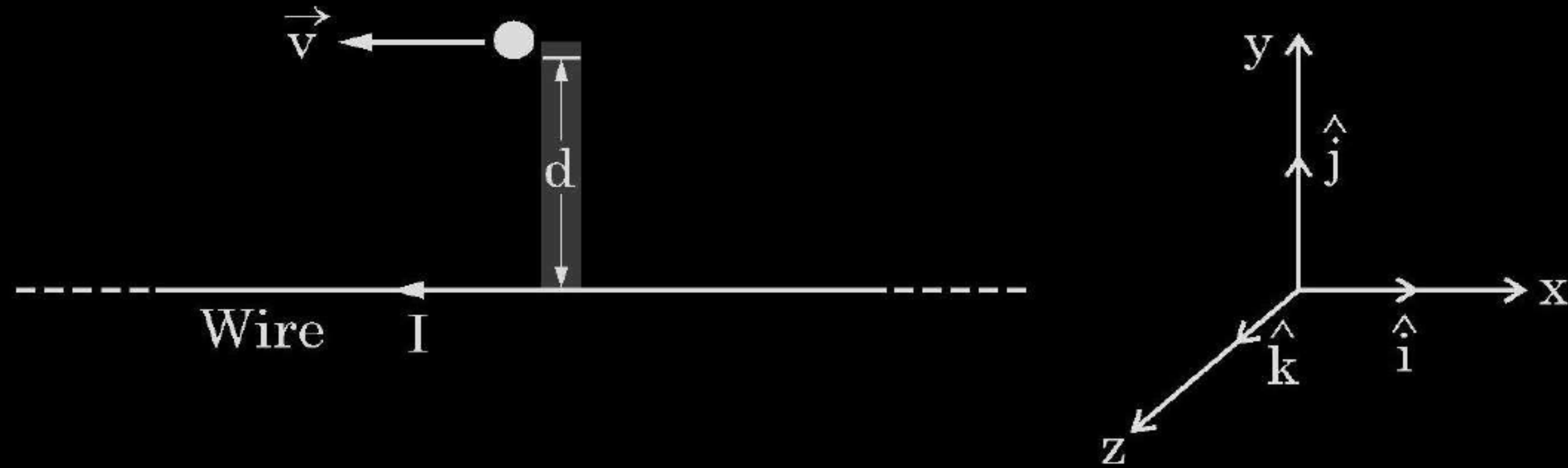
(C) $(6\hat{j} + 4\hat{k}) \text{ mN}$

(D) $(-4\hat{j} + 6\hat{k}) \mu\text{N}$

16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**
- Reason (R)** : The coil of a galvanometer is suspended in a uniform radial magnetic field.

23. A particle of charge q is moving with a velocity $\vec{v}$ at a distance ' d ' from a long straight wire carrying a current ' I ' as shown in figure. At this instant, it is subjected to a uniform electric field $\vec{E}$ such that the particle keeps moving undeviated. In terms of unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$, find –

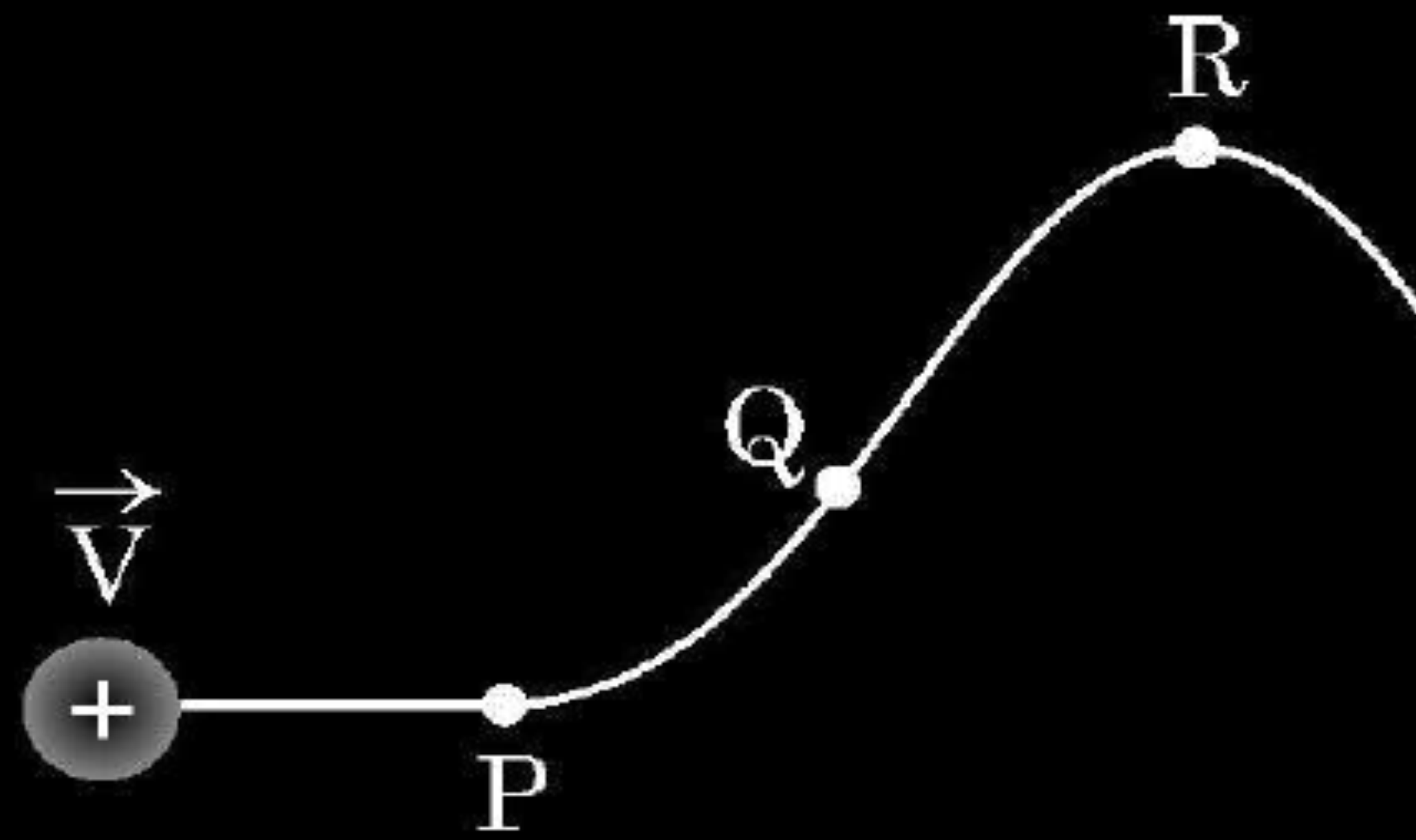
3



- (a) the magnetic field $\vec{B}$,
- (b) the magnetic force $\vec{F}_m$, and
- (c) the electric field $\vec{E}$, acting on the charge.

33. (a) (i) A proton moving with velocity $\vec{V}$ in a non-uniform magnetic field traces a path as shown in the figure.

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The path followed by the proton is always in the plane of the paper. What is the direction of the magnetic field in the region near points P, Q and R ? What can you say about relative magnitude of magnetic fields at these points ?

- (ii) A current carrying circular loop of area A produces a magnetic field B at its centre. Show that the magnetic moment of the loop

is $\frac{2 BA}{\mu_0} \sqrt{\frac{A}{\pi}}$.

- (b) (i) Derive an expression for the torque acting on a rectangular current loop suspended in a uniform magnetic field.
- (ii) A charged particle is moving in a circular path with velocity $\vec{V}$ in a uniform magnetic field $\vec{B}$. It is made to pass through a sheet of lead and as a consequence, it loses one half of its kinetic energy without change in its direction. How will (1) the radius of its path (2) its time period of revolution change ?

23. (a) Define magnetic moment of a current-carrying coil. Write its SI unit.
- (b) A coil of 60 turns and area $1.5 \times 10^{-3} \text{ m}^2$ carrying 2A current lies in a vertical plane. It experiences a torque of 0.12 Nm when placed in a uniform horizontal magnetic field. The torque acting on the coil changes to 0.05 Nm after the coil is rotated about its diameter by 90° , in the magnetic field. Find the magnitude of the magnetic field.

23. Using Biot-Savart law, derive expression for the magnetic field ($\vec{B}$) due to a circular current carrying loop at a point on its axis and hence at its centre.